

0DTEs: Trading, Gamma Risk and Volatility Propagation^{*}

Chukwuma Dim[†]

Bjørn Eraker[‡]

Grigory Vilkov[§]

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Abstract

We study the recent explosion in trading of same-day expiry (0DTE) options on the S&P500 index and examine if this trading activity is destabilizing for the underlying index. We find that Market Makers' inventory, as measured by the net gamma of their positions, is on average positive and *negatively* related to future intraday volatility. We also show evidence suggesting that positive (negative) Market Makers' inventory gamma strengthens intraday price reversal (momentum). Our empirical evidence is consistent with delta-hedging but inconsistent with information-based trading.

Keywords: 0DTE, ultra-short-term options, variance risk premium, volatility trading, gamma risk, volatility propagation

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[†]George Washington University; E-mail: cdim@gwu.edu.

[‡]Wisconsin School of Business; E-mail: bjorn.eraker@wisc.edu.

[§]Frankfurt School of Finance & Management; E-mail: vilkov@vilkov.net.

1 Introduction

The volume of short-dated options trading, especially zero-day-to-expiry (0DTE), has exploded in recent years. For the S&P500 index alone, 0DTEs accounted for 50% of the index options volume in August 2023, up from just 5% in 2016.¹ The major trading hub for equity options, Chicago Board Options Exchange (Cboe), has sequentially increased the number of weekly index option expiration dates to three in 2016 and five in 2022 to eventually have options that expire every day of the week for the next four weeks. The leading global derivatives exchange CME Group now offers 0DTEs on the aggregate market (both S&P500 and NASDAQ100) futures.

The surge in 0DTE option trading has raised several concerns among market participants and stimulated heated discussions in the press.² The primary concern is that large open positions and trading flows in 0DTEs may induce delta-hedging needs that can destabilize the underlying market, even though the underlying instruments are very liquid as in the case of the S&P500 index-based exchange-traded fund (ETF) SPY and S&P500 E-Mini (ES) Futures.³

The rationale behind these concerns, which has been modeled theoretically (e.g., Jarrow 1994, Sircar and Papanicolaou 1998, Schönbucher and Wilmott 2000), is that if option sellers delta-hedge, they trade in the direction of realized returns, e.g., sell additional shares of the underlying asset during a market decline, propagating the return shocks and strengthening momentum. The intensity with which delta hedgers re-adjust their positions in the underlying is higher for short-maturity options with strikes close to the current price level because these options' delta is more sensitive (measured by "gamma," which is inversely related to option time to expiry) to changes in the underlying. Thus, if market makers hold large short (negative gamma) inventory in 0DTEs and systematically delta-hedge it, their hedging flows can aggravate sudden market

¹E.g., [Volatility Insights: Much Ado About 0DTEs-Evaluating the Market Impact of SPX 0DTE Options](#).

²See, for example, ["Surge in zero-day options sparks fears over market volatility,"](#) or ["Short-term investors in SPY and QQQ warned of options risks."](#)

³Average daily trading volume in these markets in 2023 was around 400 bn USD.

moves. Cboe, who has observed the largest trading flows in S&P500 index 0DTEs, disputes such scenarios, claiming that these flows and the resulting market makers' exposure during the day are well-balanced and do not threaten market stability. Moreover, when the market makers' intraday net gamma is positive, delta-hedging generates trades against the realized return direction, decreasing return persistence and potentially dampening future volatility.

This paper investigates these contrasting arguments to understand the effects of the 0DTEs' trading activity on the underlying markets. We test two main channels through which 0DTE trading could impact or be linked to underlying trading and return dynamics. First, we examine whether market makers' delta-hedging needs and their potential execution in the underlying markets affect subsequent volatility and return of the underlying (gamma channel). Next, we examine an information-based channel in which market participants with superior information about near-term future volatility trade in 0DTEs prior to changes in volatility. Both channels can create the same empirically observed patterns, but the gamma channel specifies a direct cause-and-effect relationship, while the information channel suggests that observed patterns arise from traders' anticipatory actions rather than market makers' hedging flows.

Using intraday data on the underlying variance and the market makers' 0DTEs net gamma, we establish a negative and significant association between the market makers' 0DTEs net gamma and the underlying index variance, consistent with market makers' delta-hedging affecting the underlying market. We establish this result in a specification that uses high-dimensional fixed effects to account for intraday market dynamics and controls for prior realizations of the underlying variance, returns, and trading volume, thereby accounting for potential 0DTEs trading based on public information. In magnitude, we document a decline in the underlying log variance by 7.3% of its standard deviation following a standard deviation (SD) increase in market makers' 0DTEs net gamma. This effect is non-negligible, as it equals 32% of the decline in un-

derlying log variance following one SD increase in market makers' net gamma for *all* longer-term options with one-day to one-month maturities.

The observed negative association between the market makers' 0DTEs net gamma and the underlying variance could result in volatility amplification if, as feared by the media and some market participants, market makers have negative 0DTEs gamma inventory. It is, however, not what our analysis suggests. For most of our sample period, market makers are predominantly long gamma in 0DTEs, implying that their delta-hedging has the opposite effect of dampening the underlying index volatility rather than amplifying it. Moreover, since there are periods in our sample where market makers are short gamma in 0DTEs, we can investigate how the economic magnitude of their potential delta-hedging effect on the underlying variance compares with that of their long gamma exposure. Indeed, there is potential for market makers' delta hedging of their negative 0DTEs inventory to amplify the underlying variance. However, the economic magnitude of such variance amplification is 65% lower than that of the potential variance attenuation arising from market makers delta-hedging their long gamma exposure.⁴

We further quantify the economic effect of the gamma channel on the underlying return persistence. We construct an intraday momentum strategy and find that its performance is strongly negatively related to market makers' 0DTE net gamma, with 10 basis points per hour difference in average returns between high and low gamma states. We do not find similar results for market makers' net gamma in longer-term options, and it indicates that the sizeable hedging needs engendered by market makers' 0DTEs exposure tend to significantly affect intraday return persistence, boosting momentum for low gamma and reversal for high gamma states.

An important consideration is whether the established results are genuinely due to market makers' delta-hedging or stem from other market patterns, such as information-based 0DTEs

⁴In a related exercise, Vasquez, Amaya, Pearson, and Garcia-Ares (2025) quantifies the maximum negative return propagation from 0DTEs conditional on the observed data. Their estimates suggest that the maximum impact of net gamma is around 3.3 percentage points in annual terms for daily volatility, well within the range of daily changes in annualized realized volatility.

trading. We conduct several tests examining this consideration. Besides controlling for the public signals informative about future underlying volatility in our baseline analysis, which helps address some endogeneity concerns, we identify a plausibly exogenous instrument for market makers' 0DTE net gamma. We then re-examine and confirm our baseline results using instrumental variable analysis. Furthermore, we directly test whether observable public signals about future intraday underlying volatility shape the evolution of market makers' 0DTEs net gamma and reject this hypothesis.

Finally, we entertain the possibility that 0DTEs trading flow, absorbed by market makers, is driven not by public but rather by private information. Hence, we assess whether market makers' 0DTEs net gamma significantly changes in the trading windows preceding the release of major unexpected and potentially market-moving news by the media, which privately informed agents could have anticipated and traded on. Again, we do not find any evidence for such a pattern. Overall, we rely on a theoretically sound causal mechanism and the rejection of plausible alternative explanations to support our empirical evidence, consistent with a causal effect of market makers' 0DTE exposure on the underlying return volatility.

In the complementary analysis, we study the underlying market dynamics conditional on abnormal trading activity in 0DTEs. We find that the distribution of intraday returns conditional on large jumps in 0DTEs volume is similar conditional on moderate and small changes in volume. Quantile regressions show that the magnitude of positive returns on average decreases after intraday 0DTE volume jumps, whereas the magnitude of negative returns is unaffected. Importantly, our analysis indicates that 0DTEs volume does not significantly propagate past underlying returns, which is unsurprising given that a sizeable portion of 0DTEs trading flow is balanced with no significant effect of the 0DTE flow delta on subsequent volatility in the gamma channel tests.

To further understand the integration of the underlying and 0DTE markets, we use structural vector autoregression to analyze joint intraday dynamics of the 0DTE and underlying assets trading volumes and the underlying variance. 0DTEs and underlying markets are rapidly becoming more integrated, with a contemporaneous correlation between intraday trading flows increasing from 0.25-0.30 before 2021 to 0.58 in 2023-24. Positive shocks to 0DTE trading volume in recent years are associated with and followed by increasingly higher trading volume in the underlying and vice versa. Despite this increasing market integration, the difference in the magnitude of the average underlying variance response to 0DTE trading across the early and later sample periods amounts to an economically negligible 0.15 standard deviations of the return variance, corroborating the earlier results based on 0DTE volume jumps.

We contribute to several strands of research. We build on the methodology of Ni, Pearson, Poteshman, and White (2021) who analyze the link between volatility and market makers' lagged inventory net gamma in stock options. They suggest that more positive inventory gamma leads to lower volatility, consistent with the delta-hedging effect. Our analysis offers important additional insights for 0DTE index options markets: delta hedging of 0DTE positions has a larger potential for market destabilization due to more frequent and extensive rebalancing than for longer-term options, even in the case of the S&P500 index options, whose underlying markets are far more liquid than individual stocks. The potential delta hedging due to 0DTEs exposure has a distinct and non-negligible effect on the underlying index variance compared to longer-term options analyzed in the earlier studies. A closely related paper by Adams, Fontaine, and Ornathanalai (2024) confirms our results by linking intraday net gamma in 0DTEs to subsequent underlying volatility. In addition, they use variations in 0DTE trading on Tuesdays and Thursdays due to public holidays to suggest a causal effect from 0DTE trading on a given day to a lower underlying volatility on the same day. Another subsequent study that complements ours, Vasquez, Amaya, Pearson, and Garcia-Ares (2025), estimates models that link market volatility

to market makers’ inventory gamma to quantify the maximum impact of 0DTE positions on volatility. They find it to be around 3.3 percentage points (pp) per annum for daily volatility, well within the range of daily changes in annualized volatility, and 6.4 pp for 30-minute intraday volatility. These results complement our findings, suggesting that when market makers hold negative gamma inventory, the associated delta-hedging flows can moderately amplify underlying volatility.

Brogaard, Han, and Won (2023) examine the impact of 0DTE trading on daily volatility and find that 0DTEs’ relative turnover positively relates to future volatility.⁵ Baltussen, Da, Lammers, and Martens (2021) link market intraday momentum to the market makers’ gamma hedging demand. Barbon and Buraschi (2020) show that autocorrelations are significantly related to the difference between call and put options’ gamma. To our knowledge, our paper is the first to systematically study the effects of gamma exposures in 0DTEs on autocorrelation and volatility of intraday returns and analyze joint dynamics of the 0DTE and underlying markets.

A growing number of papers study patterns in 0DTEs trading (Beckmeyer, Branger, and Gayda 2023), work on specially designed pricing models for short-term options (Bandi, Fusari, and Renò 2023), and document stylized asset pricing facts about 0DTEs and ultra short-term options (Almeida, Freire, and Hizmeri 2023, Vilkov 2023, Johannes, Kaeck, Seeger, and Shah 2024). Some papers document similar results for longer-term options. Anderegg, Ulmann, and Sornette (2022) suggest that exchange rate volatility increases with the aggregate option gamma, such that delta hedgers’ order flow leads to a 0.7% (0.9%) increase in EUR/USD (USD/JPY) annualized volatility. Sornette, Ulmann, and Wehrli (2022) study Gamestop’s stock price in Spring 2021, noting that as the stock price rose, call option sellers were forced to buy, leading to a price spiral. Lipson, Tomio, and Zhang (2023) show that shocks to retail option trading due to Robinhood’s introduction of options increased optionable stocks’ return volatility.

⁵We compare the results and reconcile some of the differences in the approaches in Section 5.

The remainder of the paper is organized as follows. Section 2 conceptualizes the links between 0DTE trading and the underlying process. Section 3 discusses the data, definition of variables and presents the summary statistics for market composition and dynamics. Section 4 formally tests the hypotheses related to the channels linking 0DTE trading and underlying volatility. Section 5 presents robustness tests and additional analysis, and Section 6 concludes. Appendix and Internet Appendix contain additional materials for the analysis.

2 0DTE Trading and the Underlying Prices

Unlike the underlying stocks or index futures, which offer investors directional exposure, options expose traders to more types of risks, including time-varying directional risk (delta), realized (gamma) and implied (vega) volatility risks, time decay (theta), and others. Depending on the type of trading business and trading objectives (for example, market making with a balanced inventory and limited directional risk or speculation with both directional and realized volatility exposure), some options market participants delta hedge their options inventory to limit their directional risks.

Due to their very short expiration time, some features of 0DTEs are notably different from those of longer-term options. In particular, 0DTEs are highly sensitive to realized volatility, measured by gamma, and for close-to-at-the-money options, this sensitivity increases almost exponentially as expiration approaches. Hence, while 0DTEs can enable targeted bets around volatility-linked events, they feature stronger motivations for delta-hedging to limit the associated risks. In turn, delta-hedging trades can impact the underlying dynamics for non-informational or non-fundamental reasons.

This non-fundamental-based impact of delta-hedging on the underlying has been derived theoretically and extensively tested empirically for stock and foreign exchange long-term options. A

key assumption is that some trader groups, primarily market makers, who absorb other traders' trading flows, maintain approximately delta-neutral inventory.⁶ In contrast to information-based trading, where traders adjust options positions in anticipation of future events, the delta-hedging channel generates buying or selling pressure in the underlying conditional on the realized return and inventory characteristics. When the delta-hedging trade direction coincides with the previous return, it leads to stronger return persistence, magnifying underlying volatility. In the opposite case, delta-hedging causes lower return autocorrelation, dampening underlying volatility. We label this pattern the *gamma* channel because it implies a negative effect of market makers' 0DTEs inventory gamma on the underlying variance.

The gamma channel creates a causal effect of 0DTE activity on the underlying, and to test it, we first illustrate the relevant mechanism and define its main drivers. Assume that the underlying index price S_t follows a diffusion process with locally constant volatility so that an option's delta $\Delta_t = \frac{\partial V}{\partial S}|_{S=S_t}$ is a function of stock price and time to expiry $\Delta_t = f(S_t, T - t)$. The delta of an options portfolio is a sum of the deltas of options in the portfolio, and in the following derivations, we directly work with the delta of an aggregate inventory of market makers. Assume that at t_0 we hold a portfolio with a desired level of market exposure (delta level). To quantify the rebalancing required to maintain the same level of exposure (ignoring for now the change in the delta due to trading flow in options) at a later point in time t_1 , apply Ito's Lemma to Δ_t :

$$d\Delta = \frac{\partial \Delta}{\partial S} dS + 0.5 \frac{\partial^2 \Delta}{\partial S^2} (dS)^2 + \frac{\partial \Delta}{\partial t} dt = S \Gamma \frac{dS}{S} + 0.5 S^2 \frac{\partial^2 \Delta}{\partial S^2} \left(\frac{dS}{S} \right)^2 + \frac{\partial \Delta}{\partial t} dt,$$

⁶Delta-hedging can be expensive, especially in 0DTEs because their gamma and, hence, the required intensity of rebalancing, are increasing sharply several hours to expiration. Following Ni, Pearson, Poteshman, and White (2021), we assume that market makers are the major group of traders realistically able to delta hedge 0DTEs.

so that a discrete change in delta following small realized return r_{t_0,t_1} over a short period of time is approximately

$$\Delta_{t_1} - \Delta_{t_0} \approx S \times \Gamma \times r_{t_0,t_1} + 0.5S^2 \times \frac{\partial^2 \Delta}{\partial S^2} \times r_{t_0,t_1}^2 + \frac{\partial \Delta}{\partial t}(t_1 - t_0). \quad (1)$$

The first term quantifies the direct gamma effect, the second term quantifies the effect of gamma change, called “speed,” and the third term quantifies delta time-decay, called “charm.” It is customary to neglect the second-order and time effects for small time intervals, and we concentrate on the direct gamma effect in the main analysis.⁷ Assuming that Δ_{t_0} is at an optimal level, the market maker needs to offset the change in dollar (or cash) delta computed as

$$\widehat{\$ \Delta} \approx S^2 \times \Gamma_{t_0} \times r_{t_0,t_1} = \$ \Gamma_{t_0} \times r_{t_0,t_1}, \quad (2)$$

where $\$ \Gamma_{t_0} = S^2 \times \Gamma_{t_0}$ is the dollar (or cash) gamma of the portfolio. Thus, the direction of a delta-hedging trade relative to a future realized return direction depends only on the sign of the portfolio dollar gamma. A delta-hedger with a positive portfolio gamma sells the underlying asset after positive returns and buys it after negative returns, thus decreasing autocorrelation in underlying returns and dampening the subsequent volatility. Delta-hedgers with negative gamma will do the opposite, propagating large jumps, increasing autocorrelation, and potentially boosting volatility. In addition to delta rebalancing of the existing inventory, market makers trade periodically in the underlying market to offset the net delta of the order flow. The effect of such offsetting trades on the underlying depends on the net delta of the aggregated order flow and the return of the underlying in the period of the trade execution. Anecdotal evidence suggests that ODTEs’ order flow is composed mainly of client trades in both directions and is relatively well-balanced in the short run, but we will formally test the significance of flow delta.

⁷In the Robustness and Extensions Section 5, we also discuss the effect of speed and charm and show that including these higher-order effects does not affect our conclusion.

While the gamma channel generates a non-fundamental effect on the underlying, there are scenarios where information-based trading in options (e.g., Easley, O'Hara, and Srinivas 1998, Pan and Poteshman 2006) can generate patterns that resemble the gamma channel. For example, suppose traders have information that future volatility will be high, and as a result, buy 0DTEs. If market makers take the opposite side of the trade, their gamma inventory will decline. If subsequent volatility turns out higher, as informed traders anticipated, it will imply a negative relationship between market makers' gamma inventory and subsequent volatility. Although such a negative relationship is due to informed options trading, it may be mistaken for market makers' delta hedging effect. On the other hand, if market makers, rather than simply supplying liquidity in the market, trade options for information reasons and are better informed than other traders, there can be a positive relationship between the market makers' inventory gamma and future volatility. We label these patterns the *information* channel and try to distinguish them from the *gamma* channel in our tests.⁸

Besides the gamma and information channels, 0DTEs trading can also affect the underlying volatility if, as described in French and Roll (1986), trading in the options market substitutes trading in the underlying. Surge in 0DTEs volume could also dampen underlying volatility if it increases liquidity in the underlying market by encouraging informed investors to trade options instead of the underlying, thereby reducing the risk that market makers in the underlying market are trading against more informed investors, ultimately lowering underlying bid-ask spreads and volatility in equilibrium (Skinner 1989). In our tests, we investigate and account for these complementary 0DTEs trading and underlying volatility links.

In sum, the foregoing discussions motivate and inform our subsequent tests, which aim to document how activity in the 0DTEs market affects the underlying index volatility.

⁸It should be noted that because our analysis concentrates on the most liquid broad index, these information-based considerations, while conceptually valid, are unlikely to obtain in reality. These considerations are much more likely to apply in the context of individual stocks given their larger scope for obtaining and exploiting private information, slower pace of incorporating information into prices, and the lower liquidity for some stocks.

3 Data Preparation and Summary Statistics

We describe data processing in Section 3.1 and variable construction in Section 3.2, and then present selected summary statistics in Section 3.3.

3.1 Data Sources

Options. We work with options data for the S&P500 index (SPX) and the SPDR S&P500 ETF Trust (SPY). Index options are European, cash-settled against the S&P500 index, either at the market close for the weekly options with root SPXW, or at the market open for the “standard” SPX options. SPX has its expiration on the third Friday of each month, and SPXW expired once per week before September 2016, then three times per week, and then the fourth and the fifth days were added by Cboe on April 18, 2022, and May 11, 2022, respectively. SPY options are American and settled physically on the evening of the expiration date. Currently, SPY options have a similar expiration schedule to index options. For both assets, there are options with expiration on each day for the next month and some longer maturities. SPY price is roughly 1/10th of the index, and the notional of one SPX option is about ten times that of an SPY contract. Small notional makes SPY options popular among retail investors, and we include it in parts of our analysis.⁹

We use three intraday data formats for the options: 30-minute bars, 10-minute Cboe open-close volume summary, and actual transactions, all from Cboe DataShop. Intraday bars from 2012 to 4/2024 include national best bid and offer, trade volumes, underlying instrument price, open interest at the beginning of each day, implied volatilities, and selected sensitivities (Greeks,

⁹One can also get exposure to S&P500 options using futures options, but due to data availability, we do not include them in the analysis, though we expect no significant differences in the results. Note also that SPX index options trading takes place exclusively on the C1 platform, while ETF (SPY) options are traded on multiple exchanges. To reconstruct open interest by trader types, one needs to observe all transactions; hence, we only use index options for the analysis of intraday open interest based on the open-close volume data.

including delta and gamma).¹⁰ Open-close volume summary data for the C1 platform from 2021 to 6/2023 provide trading volumes for option contracts traded on C1, split into aggregate buying and selling flows classified by trader types: market makers, customers, pro-customers, firms, and broker-dealers. Transactions data from 2012 to 4/2024 represent enhanced Options Price Reporting Authority data. It includes trade price and size, the exchange where the trade printed, the underlying bid-ask, implied volatility, and the trade delta. The provider does not clean this data, and we parse it using filters and procedures similar to Bryzgalova, Pavlova, and Sikorskaya (2023).¹¹ After parsing, we merge simultaneous trades with the same conditions and then aggregate the data to the 1-minute bars.

In different tests, we use options on the S&P500 with both morning (AM) and regular (PM) expiration schedules (roots SPX, SPXW, and SPY) expiring within the next 30 calendar days. We compute each option’s days-to-expiry (DTE) on each day, counting only working days, and concentrate our analysis on options with extremely short maturities. Analysis based on open interest split by trader types uses only SPXW (for all maturities) and SPX (for non-0DTEs only) options.

Underlying Markets. We obtain the end-of-the-day (EOD) closing prices and 1-minute OHLC (open, high, low, close) price bars for SPX index and OHLC with traded volume for

¹⁰We recompute implied volatilities and Greeks, including speed and charm, for deep in- and far out-the-money options when they are missing in data. We exclude seven dates (24, 26/10, 14/11, and 10, 12, 19, and 24/12 of 2018) from the analysis due to missing open interest numbers.

¹¹We appreciate having access to the paper replication package at <https://tinyurl.com/reppackage>, and port the original R code to Python with slight adjustments. We keep the trades with zero `canceled_trade_condition_id`, positive `trade_size` and `trade_price`, non-negative bid-ask spread `spread`, `underlying_bid` weakly larger than 0.01 (vs. 0.1 in Bryzgalova et al.), and `trade_price` between `best_bid - spread` and `best_ask + spread`.

SPY and the continuous front contract of the S&P500 Mini-futures (ES) from *DTN IQFeed*.¹²

We add up their volumes to get the underlying volume.

3.2 Construction of the Main Variables

Realized Variances and Intraday Return Distributions. We compute intraday returns for different time intervals by aggregating 1-minute realized SPX returns. Realized variances (RV) for intraday time intervals from a given point in time t_0 to a subsequent point t_1 on the same day equal to the sum of squared 1-minute SPX log returns:

$$RV_{t_0, t_1} = \sum_{t=t_0}^{t_1-1} r_{t, t+1}^2. \quad (3)$$

Trading Activity and Risk Variables. To analyze the dynamics of market activity and aggregate risks in the options markets, we use intraday open interest by trader types computed from the Cboe open-close volume summary data, start-of-day aggregate open interest as reported for each contract in bar data, and trading volume computed from both 30-minute bar data and actual transactions (for higher frequency). Volumes and open interest are converted to dollar terms, either dollar notional or cash Greeks.

Start-of-day open interest is aggregated over a set of contracts C (combination of option roots $j \in (SPX, SPXW, SPY)$, option type, and days to expiry), and expressed in terms of

¹²In the first half of 2023, daily trading volumes in SPY and ES front contract were around 100 million shares and 1.5-2 million contracts per day, respectively. Futures have a notional of 50 units of the index, and SPY is approximately 1/10 of the index; hence, turnover in SPY and ES corresponds to 10 and 75-100 million units of the index, respectively. Thus, SPY is approximately ten times less liquid than ES, and the latter has the advantage of overnight trading sessions. Anecdotal evidence suggests that delta-hedging of index options happens in both markets, but large traders prefer Minis.

dollar notional, cash (dollar) delta and gamma, respectively:

$$\begin{aligned} OI_d^{\$} &= 100 \times \sum_{c \in C} OI_{d,c} \times S_{j,d} \\ OI_d^{\Delta} &= 100 \times \sum_{c \in C} OI_{d,c} \times \$\Delta_{d,c} \\ OI_d^{\Gamma} &= 100 \times \sum_{c \in C} OI_{d,c} \times \$\Gamma_{d,c}, \end{aligned} \tag{4}$$

where cash delta $\Delta_{d,c} = \Delta_{d,c} \times S_{j,d}$, cash gamma $\Gamma_{d,c} = \Gamma_{d,c} \times S_{j,d}^2$, with the underlying price $S_{j,d}$ are measured at the close of the first available bar on day d .

Intraday open interest by origin (firm, broker-dealer, market maker, customer, and pro-customer) is computed from open-close volume data with intraday cumulative trading volumes by contracts, split into buys and sells for market makers (MM), and buy and sell open (O), and buy and sell close (C) for the other traders. For each contract c with roots SPX and SPXW, and trader type tp , at each time t on day d , we compute cumulative order imbalance from the start of a day. For market makers, it equals to the difference between buys and sells:

$$CDOI_{d,t,MM,c} = Buy_{d,t,MM,c} - Sell_{d,t,MM,c}. \tag{5}$$

For the other trader types tp it also accounts explicitly for opening and closing flows $CDOI_{d,t,tp,c} = BuyO_{d,t,tp,c} + BuyC_{d,t,tp,c} - SellO_{d,t,tp,c} - SellC_{d,t,tp,c}$. Then we convert cumulative order imbalance to the intraday order imbalance for all time intervals $\tau = 30$ minutes as the first difference of $CDOI$ within each day: $DOI_{d,t,tp,c} = CDOI_{d,t,tp,c} - CDOI_{d,t-\tau,tp,c}$, with $DOI_{d,0,tp,c} = CDOI_{d,0,tp,c}$ to account for pre-market trading. By default, market makers absorb the net demand and supply from other market participants when managing their inventory of option contracts, and their order imbalance is equal to the aggregate order imbalance of other traders. Accumulating order imbalances $DOI_{d,t,tp,c}$ over a long period up to each point d, t gives the open interest $OI_{tp,d,t,c}$ held by tp in contract c at d, t .¹³ We compute open interest $OI_{tp,d,t,c}$

¹³Note that when there is no trading in contract c during day d , order imbalance is missing in the data and open interest can be computed incorrectly. We track and fill in zero order imbalances for all contracts that exist but are not traded on a given day. We aggregate order imbalances starting from 180 days before contract expiration.

at 30-minute intervals within each day to match bars data, and then we convert it to aggregated notional, delta, and gamma terms following the procedure in (4) and using prices and cash Greeks for the matching time points. For testing, we are specifically interested in the net gamma exposure of market makers, denoted $net\Gamma_{d,t}$, and the delta of their order flow, denoted $flow\Delta_{d,t}$, for a set of contracts C :

$$\begin{aligned} net\Gamma_{d,t} &= 100 \times \sum_{c \in C} OI_{d,t,MM,c} \times \$\Gamma_{d,t,c}, \\ flow\Delta_{d,t} &= 100 \times \sum_{c \in C} DOI_{d,t,MM,c} \times \$\Delta_{d,t,c}. \end{aligned} \quad (6)$$

Volume for a given contract c and a particular time interval $t - \tau$ to t on the day d (typically, one- or 30-minute bar) is converted to absolute cash delta terms using delta value and stock price at the end of an interval:

$$Volume_{d,t,c} = Volume_{d,t,c}^{\#} \times 100 \times |\$ \Delta_{d,t,c}|, \quad (7)$$

where $Volume_{d,t,c}^{\#}$ is the volume for contract c in number of contracts, $\$ \Delta_{d,t,c}$ is the cash delta, and 100 is the contract notional. We also define underlying trading volume for intraday time intervals $t - \tau$ to t by adding up volumes in the nearest futures contract (ES) and SPY:

$$UndVolume_{d,t} = \sum_{j \in (ES, SPY)} UndVolume_{d,t,j}^{\#} \times Z_j \times S_{j,d,t}, \quad (8)$$

where $UndVolume_{d,t,j}^{\#}$ is the volume for underlying j in number of contracts, Z_j is the notional of one contract (50 for ES, and 1 for SPY), and $S_{j,d,t}$ is the underlying price.

3.3 Market Dynamics: Summary

Trading volumes and aggregate risks. We examine the aggregate market statistics for options maturing in the next calendar month, splitting maturities into 0, 1, 2-5, 6-10, and 11-22 DTE buckets. To compute the open interest and volume, we add the respective variables for

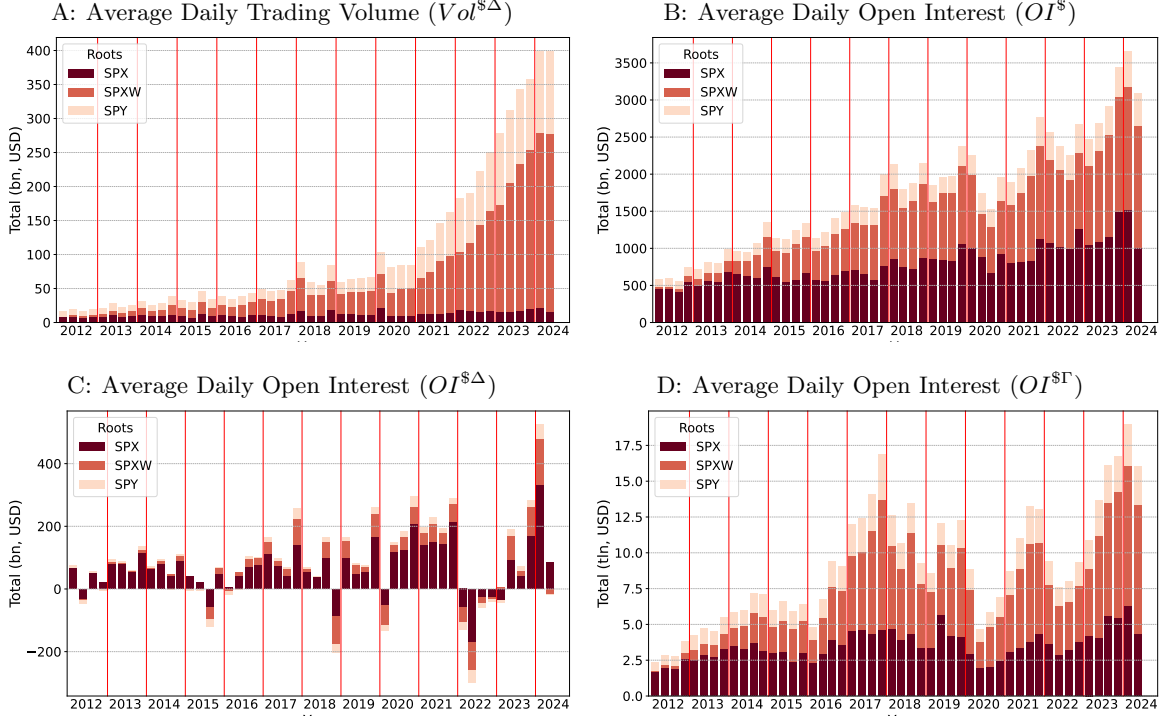


Figure 1: Aggregate Market Statistics. The figure shows quarterly averages of daily volume and open interest for SPX, SPXW, and SPY options expiring within the next 22 trading days. We aggregate variables on each day by summing over contracts up to 22 DTEs and moneyness in $[0.5, 1.5]$ the start-of-day open interest variables and the sum of intraday volume variables for all available bars, and compute quarterly averages. The sample period is from 01/2012 to 30/04/2024.

all contracts with up to 50% in- and out-the-money in the required dte range on each day d .

Figure 1 gives the first impression of the market development over the last decade by aggregating volume and open interest variables for the index and SPY options with maturities within the next month. SPY options are popular among individual investors due to the low notional value of the contract. Accordingly, the average trading volume in Panel A for SPY is relatively high—about 30% of trading in index options. At the same time, Panels B to D demonstrate that aggregate risks are predominantly held in the index options: open interest in terms of dollar notional, delta, and gamma for them is about six times higher than for the SPY. It indirectly suggests a more speculative and retail character of SPY options. Figure 2 shows volume and open interest by DTE buckets and delivers two messages: First, average daily trading volume grows mainly due to the 0DTEs; second, the open interest dollar delta (i.e., directional risk) in Panel C due

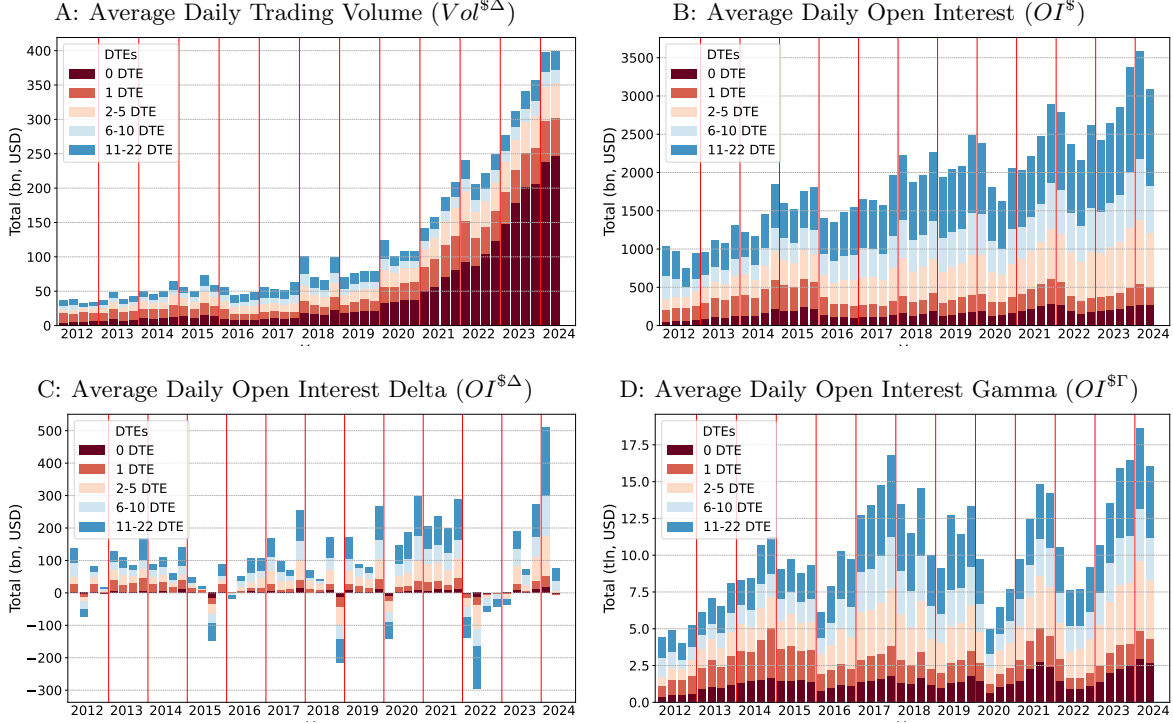


Figure 2: Market Statistics by Days to Expiration. The figure shows quarterly averages of daily volume and open interest variables for SPX, SPXW, and SPY options by expiry buckets (0, 1, 2-5, 6-10, 11-22 DTEs). We aggregate variables each day by summing over all contracts with moneyness in $[0.5, 1.5]$ in DTE buckets, including the start-of-day open interest variables and the sum of intraday volume variables for all available bars, and then compute quarterly averages. The sample period is from 01/2012 to 30/04/2024.

to 0DTEs is an order of magnitude smaller than for longer-term options and has not increased over time. The open interest in dollar notional and dollar gamma has been relatively stable over the years and is much smaller for short-term maturities. The pattern fits well with the intuition that the direction of daily risk in extremely short-term options often changes such that, on average, the dollar deltas net out. On the other hand, flipping (often) the delta of a longer-term option portfolio is costly, and its sign generally corresponds to longer-term market sentiment in a given quarter. 0DTEs' open interest and dollar gamma are sizeable, indicating a non-trivial magnitude of daily bets and resulting gamma risk on the market. Interestingly, introducing two extra weekly expiration dates in August 2016 and two more in April-May 2022 does not seem to have a pronounced effect on any of the quantities. The trading volume started a relatively sharp

increase in 2020, possibly linked to at-home COVID-related trading by individual investors or some brokers lowering options trading fees for retail customers.

Appendix Table A1 stresses the specialty of 0DTEs and the importance of analyzing ultra-short-term options separately from traditionally studied longer-term options: on average, the turnover (in terms of absolute cash delta) in 0DTEs is twice that of 1DTEs and more than four times that of all options with two to four weeks to expiry. Open interest cash gamma for 0DTEs, measuring sensitivity to realized variance and required delta-hedging intensity, is half that for *all options with one to four weeks to expiry*.

Trading activity by trader types. Market makers face trading flows from other traders and manage their options inventory to keep risks under control. Exposed to asymmetric information, they are the most probable trader group systematically delta-hedging their positions, at least partially (see, Hu 2014, Ni, Pearson, Poteshman, and White 2021). Figure 3 shows the time

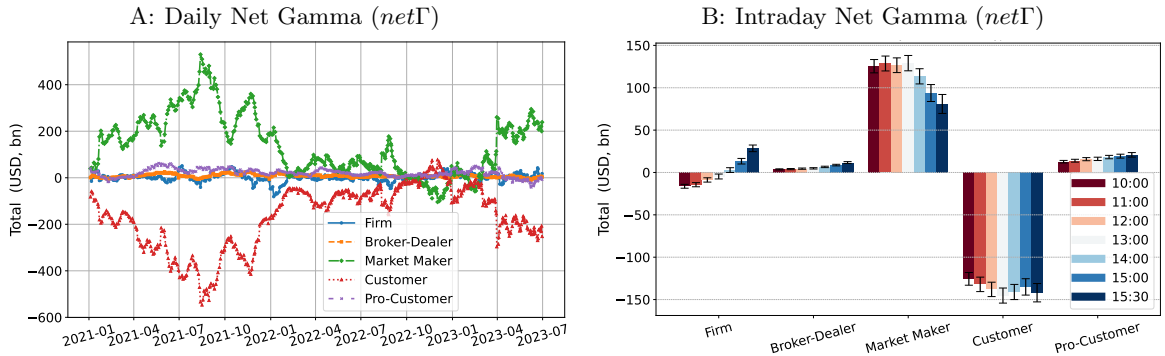


Figure 3: 0DTE Open Interest Gamma by Trader Types. The figure shows the time series of the average intraday 0DTE open interest dollar gamma $net\Gamma_{tp}$ by trader types (Panel A) and the average intraday 0DTE net cash gamma $net\Gamma_{tp}$ by points in time and trader types and their 95% confidence bounds (Panel B). Net cash gammas are computed only for 0DTE SPXW options. Series in Panel A are smoothed using an exponential moving average with a half-life of five days. The sample period is from 01/2021 to 14/06/2023.

series of average daily (in Panel A) and intraday (in Panel B) net cash gamma by trader types.

Surprisingly, market makers' net gamma stays positive for most of the sample period.¹⁴ On

¹⁴During our sample period, about one-third of observations have a negative net gamma, with an average magnitude roughly three times smaller than that of the positive gamma.

the opposite side, the major option sellers are non-professional customers, while the others hold far smaller exposures. During the day, market makers' gamma is positive and decreases monotonically towards expiry time, while customers' gamma stays roughly constant. Figure A1 shows the projected 30-minute delta-hedging flows (as a proportion of underlying volume) based on the market makers' intraday cash $net\Gamma$ values and the subsequent realized index returns: both average and maximum projected hedging flows are larger for 0DTEs compared to longer-term buckets containing options from one day (1 DTE) to expiry to a range of expiration dates (11-22 DTE). Projected hedging flows routinely exceeding 5% of the intraday volume can have a sizeable impact on the underlying and are therefore worth analyzing.

Overall, similar to the aggregate statistics, we do not observe any connection between the rapidly increasing 0DTE volume and the market makers' net gamma throughout the day.¹⁵

4 0DTE Trading and Underlying Process: Analysis

This Section specifies and formally tests the links between the 0DTE trading and the underlying process discussed in Section 2. Section 4.1 investigates the delta-hedging effect, i.e., the gamma channel. Section 4.2 analyzes alternative explanations, including the potential endogeneity of market makers' 0DTE net gamma and the competing information-based 0DTE trading channel. Section 4.3 looks into trading flows, returns, and the joint dynamics of these variables.

4.1 Shock Propagation through Delta-hedging: the Gamma Channel

We now analyze how the market makers' intraday open interest gamma relates to the future intraday realized variance, thereby shedding light on the impact of the 0DTEs market on the underlying process dynamics.

¹⁵Figure A2 in the Appendix provides additional figures for options in the other DTE buckets, which confirm that at least in the considered period, market makers are also long gamma in other maturity buckets considered in our analysis, with customers being on the other side of the open interest.

As discussed in Section 2, a direct causal mechanism through which 0DTEs can affect the underlying process is the market makers' delta-hedging activity, resulting in a negative association between the market makers' current inventory gamma and subsequent underlying variance. It follows from equation (2) that in states when market makers' gamma is positive, the delta-hedging flows push prices in the direction opposite to the realized return, thus reducing return persistence and realized variance; in negative gamma states, the effect is accordingly reversed.

To empirically assess whether the relationship between the intraday market makers' inventory in 0DTEs and the subsequently realized variance is consistent with the gamma channel, we regress the 30-minute realized log-variance following every 30-minute intraday interval up to 15:30 on the levels of market makers' 0DTEs net gamma ($0DTE\ net\Gamma_t$) observed at the end of such intervals, also accounting for exposures to longer maturities ($Non-0DTE\ net\Gamma_t$):

$$\ln RV_{t+1} = b_0 + b_1 0DTE\ net\Gamma_t + b_2 Non-0DTE\ net\Gamma_t + \mathbf{C}\mathbf{X} + \varepsilon_{t+1}, \quad (9)$$

where $\mathbf{X}$ is a vector of controls including three lags of the dependent variable, intraday returns, and changes of underlying volume, respectively, as well as fixed effects including date and a combination of year, day-of-week, and time. The dependent and continuous independent variables are scaled to unit variance for interpretability.¹⁶

Table 1 shows the estimation results, indicating that market makers' net cash gamma is negatively related to future realized variance for both 0DTEs and longer-expiry options, separately and jointly. Controls confirm an apparent persistence of realized variance, asymmetric volatility effect, and high volatility on volume effect. In terms of magnitude, columns (5) and (6) indicate a decline in the underlying index variance by 8.5% and 7.3%, respectively, relative to its standard deviation following one standard deviation increase in the market makers' 0DTE

¹⁶Note that in a simple in-sample ordinary least squares regression such a scaling does not affect either the sign or the significance of the coefficients, and it does not introduce a look-ahead bias, even though we use future information for scaling.

	$\ln RV_{t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$0DTE\ net\Gamma_t$	-0.097*** (0.011)	-0.074*** (0.018)			-0.085*** (0.010)	-0.073*** (0.018)
$Non-0DTE\ net\Gamma_t$			-0.107*** (0.008)	-0.312*** (0.044)	-0.100*** (0.009)	-0.229*** (0.046)
$\ln RV_t$	0.466*** (0.031)	0.254*** (0.024)	0.435*** (0.027)	0.226*** (0.021)	0.445*** (0.030)	0.247*** (0.024)
Ret_t	-0.050*** (0.009)	-0.043*** (0.007)	-0.058*** (0.007)	-0.045*** (0.006)	-0.048*** (0.008)	-0.040*** (0.007)
$\Delta UndVolume_t$	0.120*** (0.017)	0.053*** (0.009)	0.130*** (0.014)	0.054*** (0.008)	0.116*** (0.016)	0.054*** (0.009)
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	No	Yes	No	Yes	No	Yes
Year $\times$ Week-Day $\times$ Time	No	Yes	No	Yes	No	Yes
R-squared Adj.	0.762	0.829	0.773	0.837	0.768	0.830
Obs.	4,910	4,910	6,150	6,150	4,910	4,910

Table 1: Net Gamma and Underlying Variance. This table reports the results of a regression of the log of intraday variance over the 30-minute interval, following every 30-minute intraday time point, on the market maker net cash gamma for 0DTEs (SPXW options) and non-0DTEs (SPX and SPXW options), denoted by $net\Gamma_t$ with a prefix. The dependent and continuous independent variables are scaled to unit variance. The regression controls for three lags of the dependent variable, past returns and changes in underlying volume, respectively, but only the first lags are shown for brevity. Standard errors in parentheses use Newey and West (1987) with five lags. The sample period is from 01/2021 to 14/06/2023.

net gamma (after controlling for longer-term options positions). To put this magnitude in perspective, notice that for the market makers' non-0DTEs exposure, which comprises one-day to one-month to expiry, a standard deviation increases in the market makers' net gamma (i.e., $Non-0DTE\ net\Gamma_t$) is associated with a decline in underlying variance by 22.9% relative to its standard deviation. Hence, the economic magnitude of the effect of market makers' hedging of their 0DTEs exposures alone is roughly 32% of the magnitude for these longer maturities (from one day to one month), put together.

Therefore, the potential impact on the underlying variance arising from market makers' 0DTEs delta-hedging can be pretty sizeable. However, the direction of this impact, whether amplification or attenuation of market volatility, depends on whether the market makers are predominantly long or short 0DTEs gamma. Because market makers mostly hold positive 0DTEs net gamma positions in our sample period, as earlier shown in Figure 3, the estimates in Table 1

imply that on average their delta-hedging significantly reduces the underlying index variance rather than amplifying it.

4.1.1 Market Makers' Net Gamma Components and Conditional Effects

An important observation regarding the market makers' ODTEs net gamma is that some of the positions in the market makers' ODTEs inventory were acquired before they became ODTEs. Therefore, an interesting insight that is not obvious from the preceding analysis is whether the negative association between $ODTE\ net\Gamma_t$ and the future underlying variance is due to hedging of positions market makers acquired on the expiration day or positions previously accumulated and held until the expiration day. Such insight is important because it sheds more light on whether intraday activities on the expiration day matter or the previous result mainly reflects the effect of older but now expiring options. We examine this by decomposing the total ODTE net gamma of market makers in Table 1 into the net gamma evaluated from an open interest at the start-of-day, denoted *Start-of-day ODTE net Γ_t* , and point-in-time net gamma coming from the accumulated order imbalance during the day, denoted *Within-day ODTE net Γ_t* .

Table 2 shows the results of regressing the underlying index variance on these components of market makers' ODTEs net gamma. In all the specifications except one, both the prior exposure and positions acquired during the expiration day matter for the gamma channel. Precisely, *Start-of-day ODTE net Γ_t* and *Within-day ODTE net Γ_t* are both negatively related to future intraday variance with similar magnitudes, and both are significant even in the specifications that include the net gammas jointly in the same regression.¹⁷ These results indicate that although a sizeable chunk of the market makers' ODTEs inventory gamma could arise from positions

¹⁷The only exception with respect to the significance is in the specification in column (2), in which the initial net gamma coefficient loses significance, likely because the high-dimensional fixed effects absorb some of the variation in *Start-of-day ODTE net Γ_t* , whose variation comes only from the portfolio gamma, given that the market makers' open interest is fixed throughout the day.

acquired before they become 0DTEs, hedging exposures accumulated precisely on the expiration day can have an important effect on the underlying variance.

	$\ln RV_{t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Start-of-day 0DTE net</i> Γ_t	-0.051*** (0.010)	-0.036 (0.022)			-0.070*** (0.011)	-0.056** (0.025)
<i>Within-day 0DTE net</i> Γ_t			-0.058*** (0.007)	-0.041*** (0.011)	-0.074*** (0.008)	-0.051*** (0.011)
$\ln RV_t$	0.484*** (0.032)	0.261*** (0.024)	0.479*** (0.032)	0.257*** (0.024)	0.466*** (0.031)	0.254*** (0.024)
Ret_t	-0.051*** (0.009)	-0.044*** (0.007)	-0.050*** (0.009)	-0.043*** (0.007)	-0.049*** (0.009)	-0.043*** (0.007)
$\Delta UndVolume_t$	0.127*** (0.017)	0.053*** (0.009)	0.124*** (0.017)	0.052*** (0.009)	0.120*** (0.017)	0.053*** (0.009)
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	No	Yes	No	Yes	No	Yes
Year $\times$ Day-of-Week FE $\times$ Time	No	Yes	No	Yes	No	Yes
R-squared Adj.	0.758	0.828	0.759	0.829	0.762	0.829
Obs.	4,910	4,910	4,910	4,910	4,910	4,910

Table 2: 0DTE Net Gamma and Underlying Variance: Pre-existing vs. New Positions. This table reports the results of a regression of log of intraday variance over the 30-minute interval, following every 30-minute time point on the market maker SPXW 0DTE net gamma decomposed into the gamma at market open (*Start-of-day 0DTE net* Γ_t) and gamma of the cumulative order imbalance during the day (*Within-day 0DTE net* Γ_t). The dependent and continuous independent variables are scaled to unit variance. The regression controls for three lags of the dependent variable, past returns and changes in underlying volume, respectively, but only the first lags are shown for brevity. Standard errors in parentheses use Newey and West (1987) with five lags. The sample period is from 01/2021 to 14/06/2023.

Figure 4 extends these results by estimating the exact specification as in column (6) of Table 2 for market makers' non-0DTE inventory. Precisely, for each non-0DTE expiration bucket (*1-5 DTE* and *6-22 DTE*), we obtain the net gamma evaluated from an open interest at the start-of-day and point-in-time net gamma coming from the accumulated order imbalance during the day and include them jointly in the regression. Finally, we plot these net gamma components' coefficient estimates and confidence intervals. The figure shows that the intraday order imbalance net gamma plays a significant role for options with up to 5 days to expiry, hinting at their relatively dynamic portfolio composition. For longer-term options, only the start-of-day inventory significantly affects the intraday underlying variance.

As discussed in Section 2, delta-hedging of new positions also creates additional trade flow in underlying enforcing or weakening the return propagation depending on the flow delta and

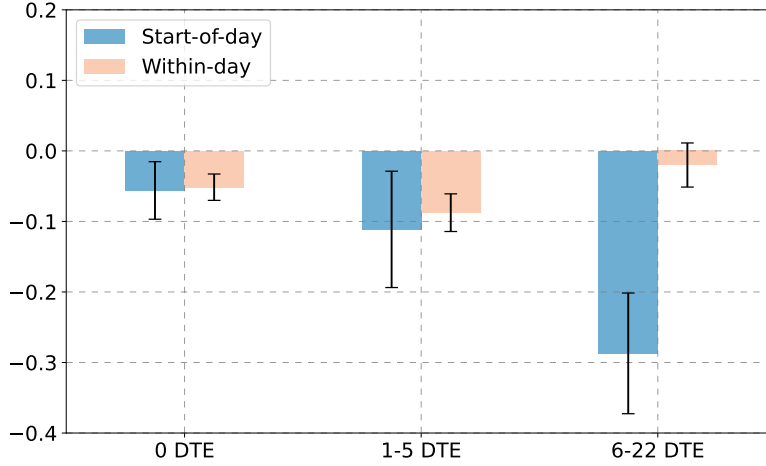


Figure 4: Net Gamma and Underlying Variance: Pre-existing vs. New Positions across DTE Buckets. The figure reports the coefficient estimates and 90% confidence interval from a regression of log of intraday variance over the 30-minute interval, following every 30-minute time point on the market maker net gamma (SPXW for 0DTE, and both SPX and SPXW for longer maturities) decomposed into the gamma at market open (*Start-of-day 0DTE netGamma_t*) and gamma of the cumulative order imbalance during the day (*Within-day 0DTE netGamma_t*), jointly for 0 DTE, 1-5 DTE, and 6-22 DTE buckets. All controls and data processing are the same as in regression (9). The sample period is from 01/2021 to 14/06/2023.

return signs. We do not find any significant effect of such initial hedging flows, and the results are provided in Column (4) of Appendix Table [IA.1](#).

Because market makers' net gamma is positive on average across days and intraday, the gamma channel evidence provided so far implies, on average, a reduction in the underlying return persistence and variance due to market makers' delta-hedging of their 0DTEs exposure. A helpful insight that complements this evidence is whether the magnitude of the effect is symmetric across scenarios where the market makers' net gamma implies an amplification vs. attenuation of the underlying variance. We, therefore, investigate the conditional effects of the market makers' 0DTEs net gamma to ascertain how different it is when the market makers' 0DTEs net gamma is positive vs. negative. Furthermore, we investigate whether the documented unconditional link materially changes when the prevailing underlying volatility is high, there is a surge in the underlying trading volume, or the underlying return is negative.

Table 3 shows in columns (1) to (3) the estimates of the following regression that includes a conditioning variable to the previous specification:

$$\ln RV_{t+1} = b_0 + ODTE \ net\Gamma_t \times (b_1 + b_2\theta_t) + \mathbf{CX} + \varepsilon_z, \quad (10)$$

where θ_t is one of the dummy variables, *High* RV_t indicating past realized variance above its sample median, *High* $\Delta UndVolume_t$ indicating a change in underlying volume above its sample median, and *Neg* Ret_t indicating a negative 30-minute return of the underlying index. Column (4) of the same table shows results for decomposing net gamma into positive and negative parts without interaction. In this specification, negative net gamma is denoted $ODTE \ net\Gamma_t^-$ and equals $ODTE \ net\Gamma_t$ but with the non-negative values set to zero. Positive net gamma is denoted $ODTE \ net\Gamma_t^+$ and equals $ODTE \ net\Gamma_t$ but with the non-positive values set to zero. Hence, by construction, $ODTE \ net\Gamma_t = ODTE \ net\Gamma_t^- + ODTE \ net\Gamma_t^+$. In all the specifications, the dependent and continuous independent variables are scaled to unit variance for interpretability.

The interactions of market makers' 0DTEs net gamma with dummies for high realized variance, surge in underlying volume, and negative realized return, respectively, are all insignificant and three to seven times smaller than the base estimate. These estimates indicate that the gamma channel effects are relatively stable over these market regimes identified using local metrics. The most important result, however, is in column (4) of Table 3: the gamma channel is strong and significant with the coefficient of -0.064 for positive net gamma, while the coefficient decreases by 65% to -0.022 for negative net gamma. This evidence suggests a much stronger underlying variance attenuation and return reversal effect coming from market makers' delta-hedging of their positive 0DTEs net gamma exposures compared to the variance amplification and higher autocorrelation generated by delta-hedging negative net gamma exposures. Overall,

	$\ln RV_{t+1}$			
	(1)	(2)	(3)	(4)
$0DTE\ net\Gamma_t$	-0.067*** (0.017)	-0.082*** (0.019)	-0.076*** (0.018)	
$0DTE\ net\Gamma_t \times High\ RV_t$	-0.028 (0.024)			
$0DTE\ net\Gamma_t \times High\ \Delta UndVolume_t$		0.014 (0.012)		
$0DTE\ net\Gamma_t \times Neg\ Ret_t$			0.011 (0.013)	
$0DTE\ net\Gamma_t^-$				-0.022** (0.010)
$0DTE\ net\Gamma_t^+$				-0.064*** (0.020)
$\ln RV_t$	0.260*** (0.026)	0.252*** (0.025)	0.250*** (0.024)	0.254*** (0.024)
Ret_t	-0.043*** (0.007)	-0.042*** (0.007)	-0.031*** (0.010)	-0.043*** (0.007)
$\Delta UndVolume_t$	0.054*** (0.009)	0.074*** (0.014)	0.054*** (0.009)	0.053*** (0.009)
Additional Controls	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Year $\times$ Day-of-Week FE $\times$ Time	Yes	Yes	Yes	Yes
R-squared Adj.	0.829	0.830	0.829	0.829
Obs.	4,910	4,910	4,910	4,910

Table 3: 0DTE Net Gamma and Underlying Variance: Conditional Effects. This table reports the results of a regression of log of intraday variance over the 30-minute interval, following every 30-minute time point on the market makers net gamma ($0DTE\ net\Gamma_t$) for 0DTE SPXW options and its interaction with dummy variables indicating last 30-minute variance of the underlying index above its sample median ($High\ RV_t$), jump in an underlying volume ($High\ \Delta UndVolume_t$), or negative last 30-minute index return ($Neg\ Ret_t$). Column (4) splits the net gamma into positive ($0DTE\ net\Gamma_t^+$) and negative ($0DTE\ net\Gamma_t^-$) values. The dependent and continuous independent variables before interaction are scaled to unit variance. The regression controls for three lags of the dependent variable, 30-minute realized returns, and underlying volumes, respectively, but only the first lags are shown. Dummies and controls' interactions are not shown. Standard errors in parentheses use Newey and West (1987) with five lags. The sample period is from 01/2021 to 14/06/2023.

the result indicates that there is indeed scope for market makers' delta-hedging of their 0DTEs exposure to amplify market shocks, but the attenuation effect dominates in our sample period.

4.1.2 Market Makers' Net Gamma and the Underlying Return Persistence

To further understand the economic magnitude of the net gamma's effect on underlying return autocorrelation, we build a simple *momentum* trading strategy similar to the one analyzed in J.P.Morgan (2023) and relate its performance to the market makers' net gamma for 0DTE and non-0DTE options. Every 30 minutes during the trading day (from 10:00 to 15:30 ET), we

compute a trading signal equal to the sign of the SPY return from the previous day's close, take a constant-size position in the underlying in the signal's direction, and close that position after sixty minutes holding period (or at 16:00 for trading at 15:30).¹⁸ We regress realized returns of the strategy on the quartiles of the 0DTE and non-0DTE net gamma at the point of position taking and date, time, and weekday fixed effects.

Figure 5, Panel A indicates that high 0DTE net gamma significantly *decreases* and low net gamma *increases* the performance of the momentum strategy, and the effects are monotonic. The evidence is consistent with the gamma channel boosting intraday return reversal for high positive net gammas and momentum for low (and negative) gammas. The difference between performance in top and bottom states is economically significant, with 10 basis points per hour (i.e., more than 0.5% per day). Non-0DTE quartiles are not significantly related to the intraday momentum; for the most quartiles the point estimates flip sign.

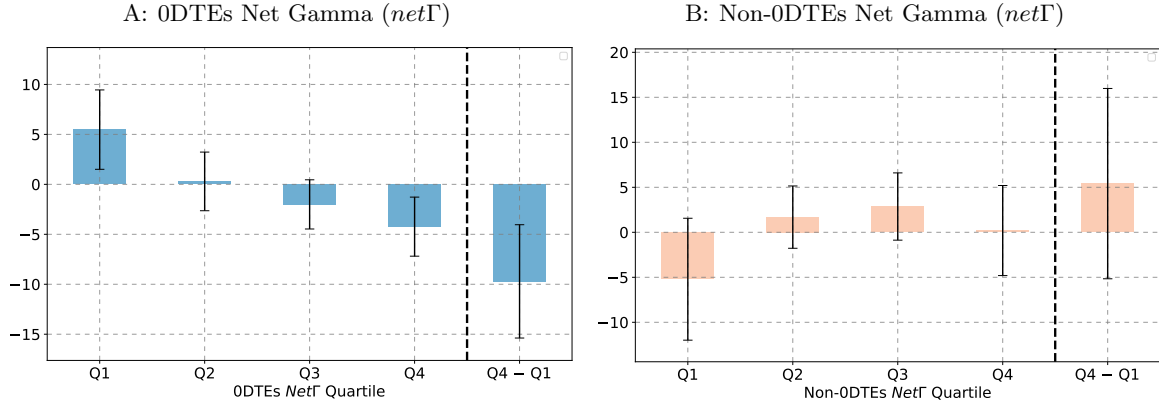


Figure 5: Mean Reversion Strategy and Market Makers' Net Gamma. The figure shows the performance of an intraday momentum trading strategy depending on the market makers' 0DTE and Non-0DTEs net gamma, respectively. Positions in SPY are taken every 30 minutes corresponding to the sign of the SPY return from the previous day's close until now and held for one hour. The resulting returns are regressed on (jointly) dummy variables for quartiles of market makers' 0DTEs (shown in Panel A) and Non-0DTEs (shown in Panel B) net gamma, accounting for date, time, and weekday fixed effects. The coefficient estimates and the 90% confidence intervals for quartiles 1 (Q1) to 4 (Q4) and the difference of Q4 and Q1 coefficients (Q4-Q1) are plotted. The confidence intervals are based on the Newey and West (1987) standard errors.

¹⁸Strategy based on trading E-Mini futures produces similar results.

4.2 Accounting for Alternative Explanations

While the observed negative link between market makers' inventory gamma and future underlying variance is consistent with the delta-hedging channel, it could also be explained by some other market patterns. For example, if market participants trade against market makers on superior information about future market variance and such information turns out to be correct, market makers' open interest gamma could be negatively related to future underlying variance even if no delta-hedging effect is in place. Therefore, we conduct tests in this section to rule out this and related considerations as the primary force driving the results documented in the previous section. In sum, our instrumental variable analysis and analyses that investigate the dynamics of market makers' open interest gamma conditional on information variables indicate that delta-hedging is the predominant force behind Section 4.1 results.

4.2.1 Instrumental Variable-based Evidence

A concern with the baseline analysis is the potential endogeneity of $\partial DTE \text{ net}\Gamma_t$ due to its possibly being driven by market activities, signals, or expectations related to future underlying variance but uncontrolled for in our analysis. We address this concern using instrumental variable (IV) regression that uses as an instrument the average of $\partial DTE \text{ net}\Gamma_t$ during the same intraday window t over the past one-week period ending five trading days (one-week) before the current day d , denoted $\overline{\partial DTE \text{ net}\Gamma}_{d-5,t}$. The instrument must satisfy the validity and exclusion criteria for the IV regression to help address the potential endogeneity concern.

For the validity criteria, the instrument has to be strongly correlated with $\partial DTE \text{ net}\Gamma_t$, and there are good reasons to expect this to be the case. The fact that we condition the instrument's construction on the intraday interval t , enables $\overline{\partial DTE \text{ net}\Gamma}_{d-5,t}$ inherit some intraday patterns in the market makers' ODTEs exposure and hence be informative about the latter. More so, because we average market makers' open interest gamma over the recent past, we capture recent

market makers' positioning in 0DTEs, which can indicate their positioning in the near future. While these intuitive properties of the instrument make it a good candidate, an instrument's strength is an empirical question that can be judged using the first stage F -statistic.

For the exclusion criteria, the instrument should not affect future underlying variance directly but only through its relationship with $0DTE\ net\Gamma_t$. The specific structure of the 0DTEs market makes a strong case for $\overline{0DTE\ net\Gamma}_{d-5,t}$ satisfying the exclusion criteria. To see this, one way to think about the endogeneity concern is that investors might have traded 0DTEs (reflected in $0DTE\ net\Gamma_t$) due to informative signals about future underlying variance. However, because 0DTE options expire on the day of trading, the horizon of the investors' expectations or signals would typically not be much longer than the 0DTEs' expiration. Otherwise, investors would have traded longer-dated options with expirations that span the investors' expectation horizon.¹⁹ As such, we assume that the information, if any, that drives 0DTEs trading is short-lived. This assumption means that trades on 0DTEs that expired several days ago prior to day $d - 5$, reflected in $\overline{0DTE\ net\Gamma}_{d-5,t}$, were unlikely based on signals about the underlying variance that will be realized at least one week in the future on day d .²⁰

We re-estimate the baseline specification that regresses log of underlying variance at 30 minutes frequency on $0DTE\ net\Gamma_t$ now using the IV procedure that uses $\overline{0DTE\ net\Gamma}_{d-5,t}$ as the instrument for $0DTE\ net\Gamma_t$, controlling for three lags of underlying variance, return and first difference of trading volume, respectively. Table 4 summarizes the estimation results, indicating that the instrument is strong across the different specifications, with the first stage F -statistic consistently above 20.²¹ We also consistently observe negative, sizeable, and strongly significant

¹⁹Reports from CBOE and other market participants confirm this view in noting that 0DTEs are suited for and typically used for targeted bets around events occurring on the 0DTEs expiration days.

²⁰The exclusion criteria may be violated in the unlikely scenario that the information about the underlying index variance that drives 0DTEs trades is long-lived and is not captured by lags of the index variables. To address this, we use data up to one week prior to construct the instrument and include three lags of the underlying variance, return, and trading volume, respectively, as controls in the IV regression. Furthermore, in subsequent analyses, we directly investigate if signals about future index variance shape $0DTE\ net\Gamma_t$, and we find that they do not.

²¹A rule of thumb is that the first-stage F -statistic should be above ten for an instrument to be considered strong enough to satisfy the relevance criteria.

coefficient estimates for the instrumented $0DTE\ net\Gamma_t$, suggesting that market makers delta-hedging of their 0DTEs exposure affects the underlying variance as hypothesized. Hence, the earlier results are unlikely due to other mechanisms besides market makers' delta-hedging.

	$\ln RV_{t+1}$				
	(1)	(2)	(3)	(4)	(5)
$0DTE\ net\Gamma_t$	-0.259*** (0.034)	-0.741*** (0.142)	-0.254** (0.112)	-0.128*** (0.040)	-0.107*** (0.040)
$\ln RV_t$	0.425*** (0.031)	0.249*** (0.035)	0.274*** (0.035)	0.489*** (0.027)	0.468*** (0.025)
Ret_t	-0.043*** (0.009)	-0.030*** (0.009)	-0.037*** (0.008)	-0.048*** (0.009)	-0.049*** (0.008)
$\Delta UndVolume_t$	0.104*** (0.016)	0.059*** (0.013)	0.042*** (0.010)	0.031*** (0.011)	0.043*** (0.011)
Additional Controls	Yes	Yes	Yes	Yes	Yes
Date FE	No	Yes	Yes	No	No
Time FE	No	No	Yes	Yes	No
Year $\times$ Week-day FE	No	No	No	Yes	No
Year $\times$ Week-day $\times$ Time FE	No	No	No	No	Yes
R-squared Adj.	0.746	0.034	0.140	0.664	0.678
Obs.	4,770	4,770	4,770	4,770	4,770
First Stage F -statistic	82.8	35.8	21.1	31.6	28.7

Table 4: Two-stage Least Squares Estimation of 0DTEs Net Gamma Effect on Underlying Variance. This table summarizes the results of the instrumental variable regressions of the log of intraday variance over the 30-minute interval, following every 30-minute intraday time point, on the market maker net cash gamma for 0DTEs (SPXW options), denoted $0DTE\ net\Gamma_t$. The average of market maker net cash gamma for 0DTEs during the same intraday window over the past one-week ending five trading days (one week) before the current day d , denoted $0DTE\ net\Gamma_{d-5,t}$, is used as the instrument for $0DTE\ net\Gamma_t$. The dependent and continuous independent variables are scaled to unit variance. The regression controls for three lags of the dependent variable, past returns and changes in underlying volume, respectively, but only the first lags are shown for brevity. Standard errors in parentheses are clustered by date. The sample period is from 01/2021 to 14/06/2023.

4.2.2 Does Public Volatility Information Drive Market Maker's Exposure?

While the preceding analysis is re-assuring, we go further and directly examine whether market makers' 0DTEs net gamma is driven by observable market variables (public information) that are likely informative about future underlying variance. We adopt the Ni, Pearson, Poteshman, and White (2021) approach and regress the future level and first difference of market makers' 0DTEs open interest gamma on three lags of the underlying variance, returns, and volume. If market makers' inventory gamma changes in response to public volatility information, we should

observe that these variables, which reflect information about future variance, significantly predict the subsequent market makers' 0DTEs open interest gamma.

The results summarized in Table 5 indicate that market makers do not significantly adjust inventory gamma in response to the public predictors of future volatility, such as past intraday volatility, realized returns, and underlying volume. While this result lends more support for

	$0DTE\ net\Gamma_{t+1}$			$\Delta 0DTE\ net\Gamma_{t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$0DTE\ net\Gamma_t$	0.754*** (0.077)	0.755*** (0.077)	0.755*** (0.077)			
$\Delta 0DTE\ net\Gamma_t$				0.016 (0.029) (0.025)	0.017 (0.029) (0.025)	0.017 (0.029) (0.025)
$\ln RV_t$	0.017 (0.023)	0.019 (0.023)	0.024 (0.031)	0.028 (0.032)	0.031 (0.032)	0.040 (0.043)
$\ln RV_{t-1}$	0.001 (0.023)	0.003 (0.023)	-0.016 (0.030)	0.007 (0.032)	0.010 (0.033)	-0.016 (0.044)
$\ln RV_{t-2}$	0.009 (0.030)	0.006 (0.030)	0.013 (0.036)	0.020 (0.045)	0.015 (0.045)	0.025 (0.054)
Ret_t		-0.005 (0.009)	-0.005 (0.009)		-0.008 (0.012)	-0.007 (0.012)
Ret_{t-1}		0.012 (0.009)	0.011 (0.009)		0.017 (0.014)	0.016 (0.014)
Ret_{t-2}		-0.000 (0.007)	0.001 (0.007)		-0.000 (0.010)	0.001 (0.010)
$\Delta UndVolume_t$			-0.005 (0.014)			-0.008 (0.020)
$\Delta UndVolume_{t-1}$			0.011 (0.017)			0.013 (0.025)
$\Delta UndVolume_{t-2}$			0.004 (0.013)			0.003 (0.019)
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Year $\times$ Day-of-Week FE $\times$ Time	Yes	Yes	Yes	Yes	Yes	Yes
R-squared Adj.	0.564	0.564	0.564	0.088	0.087	0.087
Obs.	4,908	4,908	4,908	4,907	4,907	4,907

Table 5: Market Makers' 0DTEs Net Gamma Conditional on Public Volatility Information. The table analyzes whether market makers' 0DTEs open interest gamma changes in response to information about future volatility. We regress the $t + 1$ level of market makers' 0DTEs net gamma ($0DTE\ net\Gamma_t$) and its first difference ($\Delta 0DTE\ net\Gamma_t$) on their respective three lags (only one lag for each is shown) and three lags of log realized variances, realized returns, and changes in underlying volume, respectively. The dependent variable and all continuous independent variables are scaled to unit variance. Newey and West (1987) standard errors with five lags are reported in parentheses. The sample period is 01/2021 to 14/06/2023.

the market maker delta-hedging channel and helps alleviate concerns regarding the exclusion restriction of the instrument used in Table 4, there is still the possibility that 0DTEs trading is driven by private rather than public information. The potential concern is that privately

informed agents acquire informative signals about future index volatility and trade ODTEs before the news with these signals is publicly released. If market makers take the opposite side of such trades, their open interest gamma will change in a way that makes it appear as though it is their hedging trades that drive the relationship between market makers' ODTEs net gamma and future volatility, whereas it is the preceding informed trades initiated by privately informed customers. We address this alternative information-based story next.

4.2.3 Accounting for ODTEs Trading based on Private Information

We analyze whether market makers' ODTEs net gamma changes before major unexpected news about firms that constitute the index or the aggregate US economy. Our focus on unexpected or unscheduled news allows us to examine news plausibly unknown to the general public until it is released, and hence, it could have been uncovered by privately informed traders who seek information that gives them a trading advantage over other investors.

Suppose the market makers' ODTEs net gamma is significantly affected by the options trading of privately informed investors. In that case, we should see a significant change in the market makers' ODTEs net gamma in the prior intraday trading windows before unexpected news events that might trigger underlying index volatility are made public by the news media. The idea is similar to the analysis in earlier works that examine whether informed agents trade individual stocks ahead of firm-specific news released to the public (Engelberg, Reed, and Ringgenberg 2012, Karpoff and Lou 2010, Christophe, Ferri, and Angel 2004). However, because we are analyzing the S&P500 index, it is unclear whether the relevant unexpected news privately informed investors might have uncovered ahead of other market participants pertains to the individual stocks that make up the index or is primarily about the aggregate economy.

We proceed by collecting two sets of unexpected news events. First, we collect unscheduled news about the individual stocks that comprise the index from Ravenpack. For each stock, we

count the number of news coverage and separately the number of negative news coverage (i.e., coverage with negative sentiment) released within each 30-minute window of the trading day. We aggregate these numbers to the index level by weighting the counts by the stocks' market capitalization as of the previous trading day and summing across stocks. This procedure yields a bottom-up measure of the intensity of news media coverage about the S&P500 index. From this, we obtain abnormal news coverage the standard way as the difference between the log of time t index-level news coverage and the log of its average value during the same intraday window over the past three months ending one month before the current trading day. The second set of unexpected news events focuses on unscheduled news releases about the aggregate US economy. We count the number of news coverage and, separately, the number of negative news coverage. As before, we obtain abnormal news coverage for either count type at each 30-minute window.

For either the bottom-up index-level news coverage or the aggregate news coverage, we identify episodes of intense media coverage that may be indicative of unexpected market-moving events that privately informed traders might have anticipated and acted upon before the current time t . We define such episodes based on whether the abnormal news coverage is above its 75th percentile and then examine if market makers' 0DTEs net gamma is significantly different in the periods leading up to the intense coverage; that is, in the pre-news period during which informed agents might have traded 0DTEs options, significantly changing market makers' inventory.²²

Table 6 summarizes the results of the analysis. Panel A focuses on the bottom-up media coverage, and Panel B focuses on the aggregate news coverage. Across the board, we consistently find no evidence that market makers' 0DTEs open interest gamma significantly changes in the intraday periods leading up to major unexpected news releases. This indicates that either privately informed agents do not systematically trade 0DTEs ahead of unexpected information releases, as observed in Engelberg, Reed, and Ringgenberg (2012) in the context of individual

²²Results for the abnormal news coverage above its 90th percentile are similar and are available upon request.

	Based on All News			Based on Negative News		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Bottom-Up News Coverage and 75th Percentile Threshold</i>						
<i>30-min BeforeBigNews</i>	-1.571 (1.383)		-1.587 (1.172)	-0.810 (1.206)		0.034 (1.177)
<i>1-hr ending 30-min BeforeBigNews</i>		-0.984 (1.250)	-1.029 (1.260)		-0.987 (1.189)	-0.985 (1.217)
R-squared Adj.	0.812	0.871	0.871	0.812	0.871	0.871
Obs.	6,524	5,518	5,518	6,524	5,518	5,518
<i>Panel B: Aggregate News Coverage and 75th Percentile Threshold</i>						
<i>30-min BeforeBigNews</i>	-0.455 (1.283)		0.858 (1.226)	-0.883 (1.340)		-1.014 (1.308)
<i>1-hr ending 30-min BeforeBigNews</i>		0.106 (1.344)	0.170 (1.381)		-0.743 (1.438)	-0.806 (1.466)
R-squared Adj.	0.812	0.871	0.871	0.812	0.871	0.871
Obs.	6,524	5,518	5,518	6,524	5,518	5,518

Table 6: Market Maker’s Inventory Gamma and Potential Trading based on Private Information.

This table shows the result of regressing $ODTE\ net\Gamma_t$ at 30-minutes frequency on dummy variables for trading windows before major unexpected index-level news releases. Major unexpected news releases are defined as periods when the index-level abnormal news coverage is above a certain percentile threshold indicated on the panel labels. *30-min BeforeBigNews* is a dummy variable that equals one during the 30-minute intraday window before a major unexpected news release. *1-hr ending 30-min BeforeBigNews* is a dummy variable that equals one during the one-hour intraday window ending 30-minutes before a major unexpected news release. For expositional convenience, $ODTE\ net\Gamma_t$ is divided by its standard deviation and multiplied by 100 before the estimations. In Panel A, the index-level abnormal news coverage is based on the bottom-up news coverage measure aggregating news coverage across the time t S&P500 index constituents. In Panel B, the index-level abnormal news coverage is based on news about the aggregate US economy. In Columns (1)-(3), the news coverage counts are based on all news (i.e., both positive and negative sentiment news), while in Columns (4)-(6), the counts focus only on the news with negative sentiment. All regressions include date and year $\times$ week-day $\times$ time fixed effects. Standard errors in parentheses are based on the Newey and West (1987) method with five lags.

stocks, or if such trades do occur, are absorbed by other 0DTEs’ customers and hence do not materially affect the market makers’ inventory. In any case, the evidence indicates that the negative relationship between market makers’ open interest gamma and future index volatility that we document is improbable due to 0DTEs trading based on private information.

The analyses in this section indicate that market makers’ delta-hedging of their 0DTEs exposure is most likely the force behind the evidence of Section 4.1. Neither public nor private information-based 0DTEs trading can explain the observed relationship, and the IV analyses suggest a sizeable effect of market makers’ delta-hedging on the underlying variance.

4.3 0DTEs Trading Activity and the Underlying Market

4.3.1 Intraday Return Dynamics

We analyze intraday market returns to assess how sudden spikes in 0DTE trading volume shape subsequent return realizations. We are interested in the accumulation of realized returns in either the positive or negative direction following a volume jump, with the negative reaction expected to have stronger implications for market stability. While high volume is often associated with high return volatility, we want to ascertain whether sharp jumps in 0DTE volume precede and are potentially related to large underlying returns in the spirit of earlier theoretical and empirical works that suggest a link between options trading volume and underlying prices and volatility (e.g., Easley, O’Hara, and Srinivas 1998, Pan and Poteshman 2006). It can be driven by a preferred trading venue for informed traders or mechanical delta-hedging effects of an unbalanced order flow—we stay agnostic about the mechanics now and analyze whether we can document such a temporal link empirically and whether in our extended sample from 2012 to 04/2024 (compared to 01/2021 to 06/2023 used in Sections 4.1 and 4.2) we observe any extreme return realizations following 0DTE volume spikes.

To obtain 0DTE options volume, we sum the dollar-delta volumes for SPY and SPXW options at each 1-minute intraday interval and then convert it to log volume $v_t^{\$ \Delta} = \ln Vol_t^{\$ \Delta}$. We identify jumps in intraday 0DTE trading volume by point in time t with the 1-minute changes in trading volume $(v_t^{\$ \Delta} - v_{t-1}^{\$ \Delta})$ larger than three times its standard deviation on a given day. We use the 5-minute cumulative returns (from $t + 1$ to $t + 5$) as a measure of market reaction to a volume shock at time t and then test whether the market reaction differs conditional on having experienced a sizeable 0DTE volume jump and whether it depends on the realized return before and during the 0DTE trading volume jump.

Figure 6 plots the distribution densities of the 5-minute cumulative returns conditional on jump vs. no jump in ODTEs volume and separately for four sub-periods. Visually, the distribution conditional on volume jumps has slightly fatter tails on both sides of the return realizations, but we do not observe drastic differences in the distributions or out-of-the-ordinary market reaction following ODTE volume jumps. We further test these observations more formally. First, we run two non-parametric tests (k -sample Anderson-Darling and two-sample Kolmogorov-Smirnov tests) to see whether both samples are drawn from the same distribution. Second, because we are especially interested in the distribution's tails, we run a series of quantile regressions analyzing how relatively infrequent realizations of returns depend on ODTE volume jumps dummy in interaction with year fixed effect and cumulative market returns before the volume jump. Such specification lets us directly evaluate claims that large underlying market moves are propagated through ODTE trading, especially in recent years.

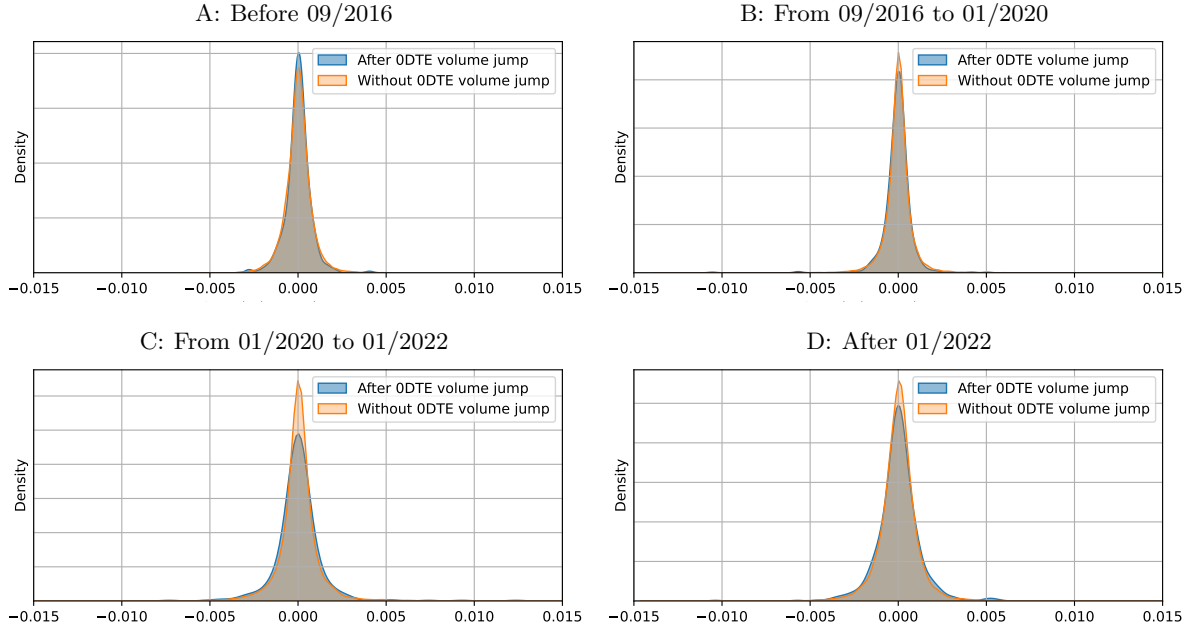


Figure 6: Cumulative Returns Conditional on ODTE Volume Jump. The figure shows the distribution density of 5-minute returns (from $t + 1$ to $t + 5$) conditional on having a large jump in ODTE volume at t and not having such a jump, for four sub-periods. We sample non-overlapping observations, skipping at least five minutes between each selected t point. The sample period is from 01/2012 to 30/04/2024 and includes only days with ODTE expiration.

Table 7 shows the results of the non-parametric tests for the conditional samples. We fail to reject that the samples are drawn from the same distribution for all sub-periods and both tests. k -sample Anderson-Darling is more suitable for our purpose because, compared to the Kolmogorov-Smirnov, it puts more weight on the tails of the distributions. Table 8 presents

	KS Statistic	p-val	AD Statistic	p-val
Full Sample	0.025	0.118	0.705	0.169
Before 09/2016	0.061	0.184	0.721	0.166
From 09/2016 to 01/2020	0.034	0.363	0.422	0.223
From 01/2020 to 01/2022	0.039	0.513	-0.692	0.250
After 01/2022	0.022	0.809	-0.358	0.250

Table 7: Testing Conditional Cumulative Return Samples. This table reports the results of the two-sample Kolmogorov-Smirnov and k -sample Anderson-Darling tests (Scholz and Stephens 1987) for the distributions of 5-minute cumulative returns (from $t + 1$ to $t + 5$) conditional on having a large jump in 0DTE volume at t and not having such a jump, for four sub-periods. We sample non-overlapping observations in the sample period from 01/2012 to 30/04/2024 and includes only days with 0DTE expiration.

the results of the quantile regressions for selected percentiles on both sides of the distribution, using the period before 2020 as the base. We include all double and triple interaction terms in the regression but concentrate on past cumulative return and 0DTE volume jump, with and without year dummy interaction. We report only the interaction term coefficients central to our analysis for brevity. In all cases, we fail to reject the null that there is no propagation of past returns by 0DTE volume jumps—the interactions of past return and volume jump, with and without year dummies, are all insignificant. There is limited evidence that the 0DTE volume jump is inversely associated with positive future returns (Q90 and Q95), but the relationship is not mediated by the past returns preceding the 0DTE volume jump. In short, there is no evidence that 0DTE trading *propagates* market moves.

4.3.2 Underlying and Options Markets Integration

Next, we examine both the underlying and options markets jointly by studying how intraday trading in both markets is related to realized price movements of the underlying asset. To this end, we estimate Vector-Auto-Regressions (VARs) on three key time-series: 0DTE options trad-

	Q1	Q5	Q10	Q90	Q95	Q99
0DTE Volume Jump	-0.006 (0.023)	0.002 (0.008)	0.003 (0.005)	-0.010** (0.005)	-0.019** (0.008)	-0.037* (0.022)
Past Return	2.037 (10.933)	1.114 (3.049)	1.029 (1.750)	-6.240*** (1.500)	-6.289** (2.912)	-4.226 (10.990)
0DTE Volume Jump x Past Return	-51.693 (104.024)	-3.653 (21.889)	-1.406 (11.904)	24.149 (14.696)	24.660 (31.023)	17.382 (110.693)
0DTE Volume Jump x Year 2021	0.016 (0.066)	-0.005 (0.020)	0.009 (0.014)	0.016 (0.012)	0.032 (0.020)	0.017 (0.065)
0DTE Volume Jump x Year 2022	-0.009 (0.050)	-0.003 (0.017)	-0.002 (0.011)	0.004 (0.010)	0.009 (0.017)	0.157*** (0.052)
0DTE Volume Jump x Year 2023	-0.052 (0.047)	-0.030* (0.017)	-0.026** (0.011)	0.029*** (0.010)	0.079*** (0.017)	0.075 (0.049)
0DTE Volume Jump x Year 2024	0.001 (0.088)	-0.006 (0.026)	0.005 (0.017)	0.004 (0.015)	0.029 (0.026)	0.089 (0.081)
Past Return x 0DTE Volume Jump x Year 2021	83.374 (262.558)	11.315 (31.844)	-2.722 (25.125)	-21.948 (28.243)	-32.356 (62.968)	-25.468 (327.930)
Past Return x 0DTE Volume Jump x Year 2022	67.084 (108.011)	2.914 (24.789)	14.479 (14.418)	-17.711 (16.309)	-22.025 (32.953)	4.772 (116.329)
Past Return x 0DTE Volume Jump x Year 2023	73.484 (115.453)	10.704 (31.885)	11.530 (16.887)	-19.975 (17.804)	10.135 (34.878)	11.449 (121.656)
Past Return x 0DTE Volume Jump x Year 2024	85.573 (167.320)	-34.659 (34.659)	-5.363 (21.556)	-28.763 (26.513)	-21.670 (54.512)	-0.418 (178.580)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	99,707	99,707	99,707	99,707	99,707	99,707

Table 8: Quantile Regressions of Intraday Returns. This table reports the results of the quantile regressions of the selected percentiles (1/5/10/90/95/99) of the five-minute cumulative returns from $t + 1$ to $t + 5$ on the dummy for the 0DTE Volume Jump at t , past five-minute cumulative return from $t - 4$ to t (Past Return), year dummies, and double and triple interactions of the variables. We omit some double interactions from the table for space reasons. We sample non-overlapping observations, skipping at least five minutes between each selected t point. The sample period is from 01/2012 to 30/04/2024 and includes only days with 0DTE expiration. Standard errors are in parentheses.

ing volume, underlying volume, and Realized Variance of the underlying. We seek to understand the joint dynamics of these variables, their contemporaneous correlation, and how the dynamic relationship between them has changed over time as 0DTE trading has become more prevalent. Because trading volumes in the underlying and options markets are interrelated and linked to realized returns through delta-hedging, we model their dynamics jointly using a structural VAR. Following Koop, Pesaran, and Potter (1996) and Pesaran and Shin (1998), we analyze responses to shocks in the system using generalized impulse response functions (gIRF), which account for the correlation of structural shocks in the system and are invariant to the ordering of the time-series (Pesaran 2015 and Wiesen and Beaumont 2024).

In order to compute volume for the underlying, we use both intraday 1-minute bar data for SPY, and the front-month contract of the S&P500 E-mini Futures, ES. At the one-minute

frequency, for each bar within the regular day session (from 9:31 to 16:00), we add up dollar trading volume $V_j^\$$ for both $j = \text{SPY}$ and ES to obtain aggregate volume $V_{d:t}^\$$ and convert it to $\log v_{d:t}^\$ = \ln V_{d:t}^\$$. For the volume of 0DTE options, we proceed in the same way, adding dollar delta volumes for each one-minute for SPY and SPXW options and then convert it to $\log v_{d:t}^{\Delta} = \ln V_{d:t}^{\Delta}$. We also compute simple SPY returns $R_{d:t}$ over the matching time intervals and use 1-minute squared returns as the instantaneous variance proxy $\sigma_{d:t}^2 = R_{d:t}^2$. To maintain stationarity in the time-series, we use the first differences of the volume variables for estimation. We also standardize all variables daily to unit variance.

Because we want to focus on the variables' intraday association, we estimate their joint dynamics each day, compute the generalized impulse response functions, and use the distribution of these gIRFs over a specified period to get the average impulse responses and their confidence bounds. We split the sample into two sub-periods: 2012-2019, characterized by relatively less 0DTE volume, and 2020-2024, during which 0DTE trading volumes rapidly increased. As expected, the trading volume differences for 0DTEs and the underlying are highly correlated, especially in the latter period: the correlation goes from 0.27 in 2012-2019 to 0.43 in 2020-2024. The correlations between the variance and volume variables are initially higher for the underlying market than for 0DTEs (0.14 vs. 0.08). However, in the latter period, both correlations almost perfectly align at the level of 0.15–0.16.

We estimate the following structural model daily as a dynamic reduced-form VAR at 1-minute frequency and using $n = 5$ lags:

$$\begin{aligned}
\sigma_{d:t}^2 &= c_0 + \sum_{l=1}^n c_{1,l} \Delta v_{d:t-l}^\$ + \sum_{l=1}^n c_{2,l} \Delta v_{d:t-l}^{\Delta} + \sum_{l=1}^n c_{3,l} \sigma_{d:t-l}^2 + e_{\sigma^2, d:t}, \\
\Delta v_{d:t}^{\Delta} &= b_0 + \sum_{l=1}^n b_{1,l} \Delta v_{d:t-l}^\$ + \sum_{l=1}^n b_{2,l} \Delta v_{d:t-l}^{\Delta} + \sum_{l=1}^n b_{3,l} \sigma_{d:t-l}^2 + e_{\Delta v^{\Delta}, d:t}, \\
\Delta v_{d:t}^\$ &= a_0 + \sum_{l=1}^n a_{1,l} \Delta v_{d:t-l}^\$ + \sum_{l=1}^n a_{2,l} \Delta v_{d:t-l}^{\Delta} + \sum_{l=1}^n a_{3,l} \sigma_{d:t-l}^2 + e_{\Delta v^{\$}, d:t}.
\end{aligned} \tag{11}$$

We use the estimation output to compute generalized impulse response functions for one-standard deviation shocks to the selected variables.²³ For each sub-period 2012-2019 and 2020-2024, we obtain the average values and the confidence bounds (5th and 95th percentiles) from the collection of intraday gIRFs for five time steps. Figure 7 shows three key impulse response

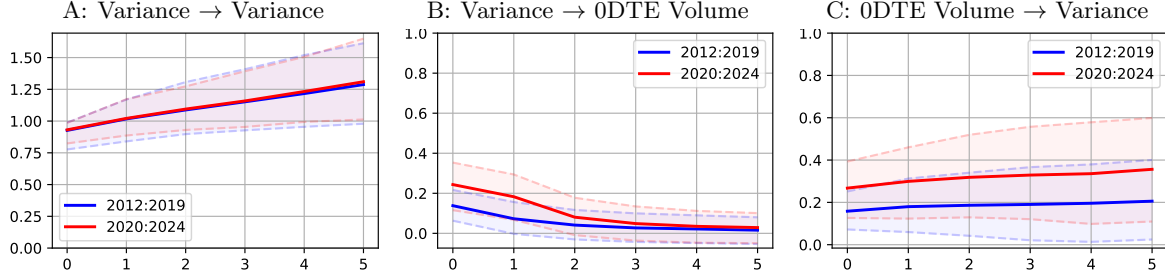


Figure 7: 0DTE Volume and Underlying Variance Generalized Cumulative Impulse Response Functions. The figure shows the average generalized impulse response functions with confidence bounds (5th and 95th percentiles of empirical daily distribution) for the VAR system in (11) estimated daily with $n = 5$ lags for one-minute frequency. The averages and percentiles are computed for 2012-2019 (plotted in blue) and 2020-2024 (plotted in red). The response is calculated for one standard deviation shock to a given variable. The variables are winsorized at 0.01/0.99 levels for the whole sample period and scaled daily to unit variance. The sample period is from 01/2012 to 30/04/2024.

functions. To interpret the plots, note that a difference between the two time periods will show as a parallel shift in the gIRFs *if* the change in the gIRF is driven by a *contemporaneous* correlation. We see this in the last plot (Panel C), where the average post-2020 gIRF is shifted upwards by an amount corresponding to the 0-frequency shift, indicating the response of market variance to shocks in 0DTE volume is entirely driven by an increase in the 0-frequency, contemporaneous correlation between these variables. We observe no change between the periods for the Variance-to-variance IRF (Panel A). Panel B shows how variance shocks impact 0DTE volume. Here, we see a slight increase at the 0, 1, and 2-minute horizon, followed by a near-zero difference at longer horizons.

Figure 7 also shows confidence bands for the gIRFs computed from their empirical distribution. We see that these bands overlap and, thus, suggest that we cannot reject the null that

²³Note that one can easily extend the analysis to compute joint impulse response functions as in Wiesen and Beaumont (2024). However, it requires an assumption about joint shocks, which adds unnecessary complexity.

the dynamic relationship implied by the VARs is unchanged from the pre- to post-2020 periods. The point estimates, however, imply a small but statistically insignificant increase in the joint correlation between the three variables at the contemporaneous, zero-frequency horizon. We interpret this to suggest an increase in market integration over time. For example, following a one standard deviation shock to 0DTE trading, the underlying index variance rises from about 0.18 standard deviation units in the 2012-2019 sub-period, increasing to 0.26 standard deviation units in 2020-2024. Although these numbers imply a 1.5 times stronger response in the latter period, the magnitude is economically negligible and statistically insignificant.

5 Robustness and Extensions

We conduct several tests to assess the robustness of our findings, reconcile some of our results with the existing literature, and collect the additional results in the Online Appendix.

We extend our analysis of market maker open interest gamma effect on underlying variance to include “speed” and “charm” in Section [IA.1.1](#), and the results indicate that these components do not propagate past volatility or adverse market shocks. We also find no significant effect of the delta of intraday 0DTE flow interacted with the realized return sign.

In Section [IA.1.2](#), we analyze the discrepancies between our results and findings of the positive effect of 0DTE trading volume on the underlying variance in Brogaard, Han, and Won (2023). Table [IA.2](#) shows the results of a regression of daily variance on lagged 0DTE volume and open interest gamma, with and without lagged variance controls. The significant positive coefficient on lagged 0DTE volume turns insignificant once we control for lags of the underlying variance. The reported results vary considerably depending on the variables pre-processing and controls (e.g., applying log transformation to the dependent variable, controlling for lags of overnight

variance and using volume ratio instead of log volume). Given this lack of robustness, we are cautious in interpreting these numbers and their statistical properties.

Taken together, these two validation exercises show that even though market makers' 0DTEs exposure can affect the underlying variance for non-fundamental reasons—in our sample the effect on average is underlying variance attenuation—0DTEs trading volume by itself does not necessarily have this same effect. One reason for this is that as Figures 2 and 3 show, high 0DTEs trading volume does not imply high market makers' exposure to 0DTEs that needs to be delta-hedged by trading in the underlying market. Instead, a significant portion of the 0DTEs volume could well be trades between other market participants, which does not change the market makers' inventory and potential hedging needs.

In Section IA.1.3, we look at the link between aggregate open interest gamma levels and daily volatility, assuming that aggregate open interest, representing a total outstanding risk of options inventory, can proxy for the delta-hedging demand. However, we do not find any significant effect in Table IA.3, confirming the importance of knowing market makers' inventory for quantifying the link between option market activity and underlying volatility.

Section IA.1.4 provides additional analysis of sensitivity of impulse response functions to 0DTEs activity in Table IA.4. It also extends the market integration analysis with VAR to longer-term maturities, showing in Figure IA.1 that the higher intraday volume and variance shocks correlation is driven predominantly by short-term options expiring within one week.

6 Conclusion

Following the introduction of weekly options with daily expiration by Cboe, ultra-short maturity options have surged in popularity in recent years, with the daily trading volume in zero days to expiration options (0DTEs) increasing more than tenfold from its 2012 level. As a result, several

large market participants and major media outlets have expressed concerns that the ballooning 0DTE trading could intensify price moves following the re-adjustment of delta hedges in the same direction as the underlying market.

Following the basic causal mechanism from inventory delta-hedging to trading flows and price reaction, we document that market makers' intraday 0DTE open interest gamma is negatively associated with future short-term volatility and persistence of intraday returns. An intraday momentum trading strategy earns 10 basis points per hour more in states with low market makers' 0DTE gamma than in high gamma states, indicating that the effects are economically significant. We do not find empirical support for the alternative information channel linking market makers' gamma and future volatility, thus confirming the validity of the causal link between the market makers' 0DTE inventory and subsequent intraday volatility through the gamma channel.

In our sample period, market makers carry a positive 0DTE net gamma on average and most days, meaning that their delta-hedging typically dampens the underlying volatility. While the gamma-to-volatility link is weaker in negative gamma states, this does not preclude the possibility of 0DTE market activity strongly propagating adverse market shocks.

We also analyze how spikes in 0DTEs trading volumes shape the subsequent intraday market index returns. We do not find significant differences between distributions on days with and without volume jumps. More so, extreme quantiles of the intraday return distributions are not significantly related to the interaction of 0DTE volume and preceding returns. Nonetheless, we observe stronger integration of the underlying and 0DTE option markets, with more correlated trading and simultaneous spikes in volumes and return variance.

In conclusion, while our results suggest that market makers' 0DTE positions normally do not propagate large market moves and are linked to a lower, not higher, short-term underlying volatility, the results are conditioned on a period when market makers, on average, held long

positions in 0DTEs. Thus, while concerns over a potential hedging-based destabilization of underlying markets are not supported by our evidence and sample period, we do not extrapolate to a scenario where the market makers' inventories become very negative. Future studies could assess the limitations of the existing 0DTEs regulatory frameworks (e.g., intraday exchange margin and mark-to-market mechanisms) and their potential effect on market stability. Such analysis could yield insights that can aid the design of optimal risk management and regulatory practices, which are beyond the scope of the current study.

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A Additional Tables and Figures

	Count	Mean	StDev	Min	25%	50%	75%	Max
<i>0 DTE</i>								
Trade Volume (USD, bn)	1685	275.3	296.0	4.1	44.1	126.0	485.3	1409.8
Trade Volume Delta (USD, bn)	1685	68.8	75.2	1.6	12.7	29.8	111.3	367.9
Open Interest (USD, bn)	1685	174.2	124.2	11.3	85.5	144.7	223.3	1130.6
Open Interest Delta (USD, bn)	1685	4.2	21.0	-135.4	-3.9	3.2	12.1	198.3
Open Interest Gamma (USD, bn)	1685	1538.5	1008.4	7.0	790.9	1312.2	2036.9	7687.2
<i>1 DTE</i>								
Trade Volume (USD, bn)	1689	113.0	89.9	2.9	40.7	84.2	171.6	658.0
Trade Volume Delta (USD, bn)	1689	28.5	24.5	0.8	9.4	20.9	42.9	228.9
Open Interest (USD, bn)	1689	216.8	333.7	7.4	61.9	102.9	228.9	3468.4
Open Interest Delta (USD, bn)	1689	11.3	70.5	-712.3	-2.7	2.5	9.9	881.2
Open Interest Gamma (USD, bn)	1689	1445.3	1876.7	48.7	486.7	872.4	1558.3	20338.6
<i>2-5 DTE</i>								
Trade Volume (USD, bn)	2910	94.4	74.0	3.0	40.6	69.3	129.8	438.7
Trade Volume Delta (USD, bn)	2910	21.0	17.1	0.5	8.7	15.1	29.3	121.5
Open Interest (USD, bn)	2910	293.6	408.4	4.2	57.0	131.1	292.3	3325.2
Open Interest Delta (USD, bn)	2910	19.3	88.7	-692.6	-2.6	2.9	14.5	907.4
Open Interest Gamma (USD, bn)	2910	1657.1	2032.9	28.0	421.0	881.4	1954.0	17834.2
<i>6-10 DTE</i>								
Trade Volume (USD, bn)	3009	45.9	32.6	0.8	23.2	38.4	60.1	267.4
Trade Volume Delta (USD, bn)	3009	10.7	8.8	0.1	4.7	8.3	14.1	82.6
Open Interest (USD, bn)	3009	277.9	418.3	3.9	29.5	100.3	381.8	3125.9
Open Interest Delta (USD, bn)	3009	18.8	82.3	-655.9	-1.8	1.6	11.4	826.2
Open Interest Gamma (USD, bn)	3009	1368.1	1923.7	10.7	202.0	536.0	1769.4	13466.5
<i>11-22 DTE</i>								
Trade Volume (USD, bn)	3053	62.3	44.5	0.6	31.7	52.5	83.9	350.9
Trade Volume Delta (USD, bn)	3053	15.2	12.1	0.1	7.0	12.0	20.4	118.6
Open Interest (USD, bn)	3053	461.4	491.8	0.6	37.7	380.2	688.2	3009.5
Open Interest Delta (USD, bn)	3053	27.7	96.0	-384.2	-1.6	2.1	40.6	742.3
Open Interest Gamma (USD, bn)	3053	1983.0	2631.3	6.7	221.5	1512.5	2882.8	93509.5

Table A1: Volume and Open Interest by DTE Buckets. The table provides statistics for the selected open interest and volume variables defined in equations (4) to (7) for all options with roots SPX, SPXW and SPY aggregated by DTE buckets (note that we exclude SPX options from 0DTE statistics, because they expire at the open of the day). The variables are first aggregated for each day: open interest variables use open interest in terms of number of contracts at the beginning of the day, and the underlying prices and option deltas and gammas reported at 10:00; trade volume variables are first computed for each 30-minute bar during the regular session from 9:30 to 16:00 using volume in contracts during each 30-minute interval, and underlying prices and deltas at the end of each bar, and then added up for each day. The sample period is from 01/2012 to 30/04/2024.

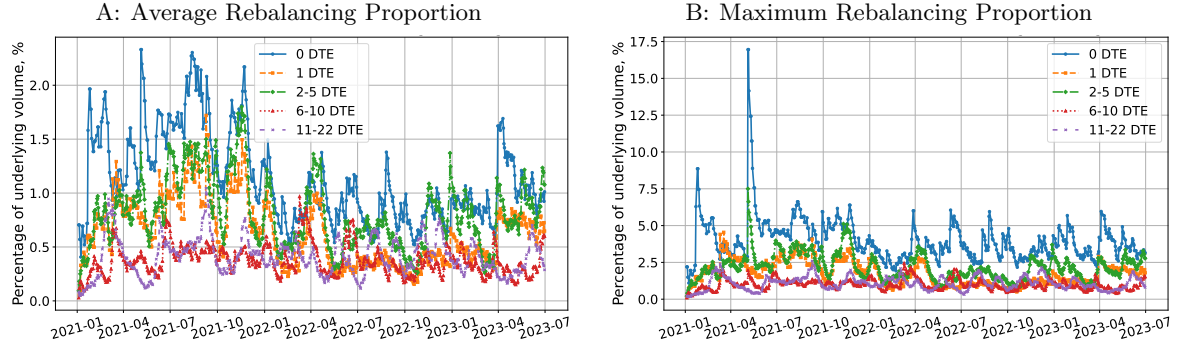


Figure A1: Net Gamma-Projected Rebalancing by Market Makers. The figure reports the required delta rebalancing by market-makers as a percentage of the total underlying volume (nearest futures, ES, and SPY combined). We approximate the required delta rebalancing volume as the product of market makers' cash $net\Gamma_t$ for each DTE bucket at the start of each 30-minute intraday bar and the absolute realized return during the next 30 minutes. We divide these volumes by the total volume in underlying during these 30 minutes to compute percentage and take the average (Panel A) or the maximum of this percentage for each day. Series are smoothed using an exponential moving average with a halflife of five days. The sample period is from 01/2021 to 14/06/2023.

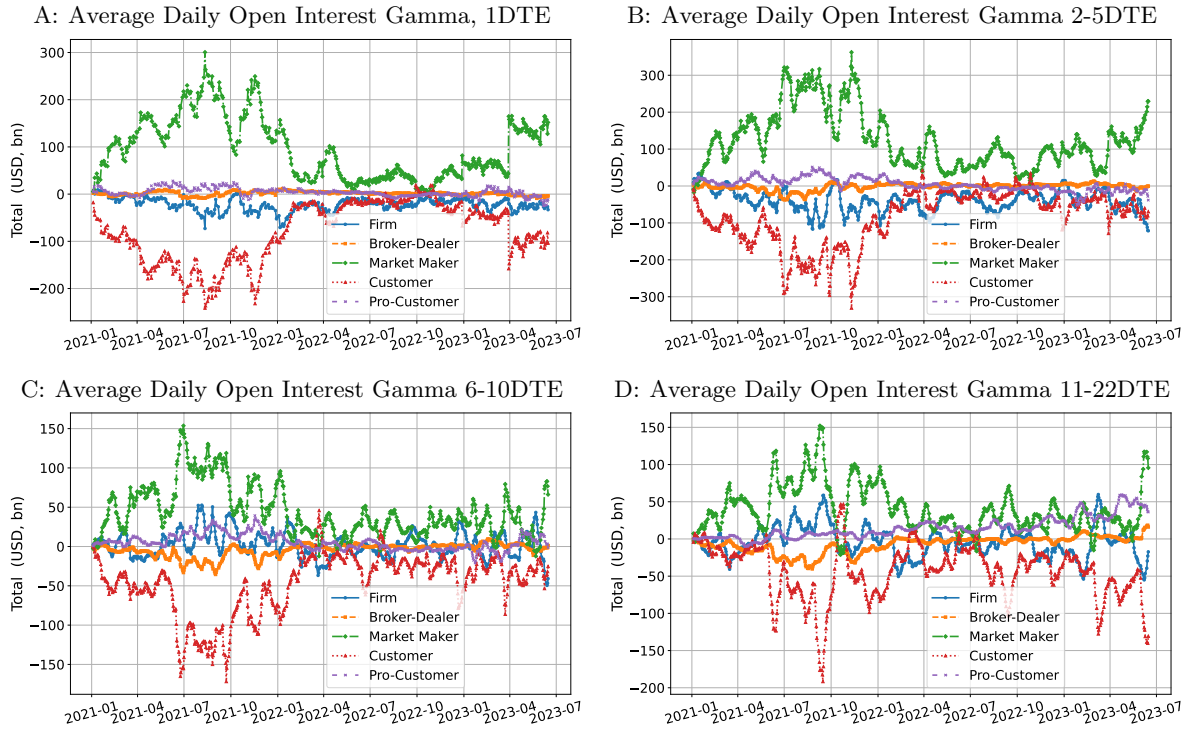


Figure A2: Open Interest Gamma by Trader Types and Days to Expiry. The figure shows the time series of the average intraday open interest dollar gamma $OI^{\$ \Gamma}$ aggregated by trader types for different days to expiry buckets (from 1DTE in Panel A to 11-22DTE in Panel D). Open interest gammas are computed for SPXW options expiring on the day of trading. Series are smoothed using exponential moving average with halflife of five days. The sample period is from 01/2021 to 14/06/2023.

Online Appendix
for
0DTEs: Trading, Gamma Risk and Volatility Propagation

This version: June 6, 2025

IA.1 Robustness and Extensions

IA.1.1 Open Interest Gamma Channel

Intraday Market Maker Open Interest Speed, Charm, and Flow Delta Effects. In Section 2, we split the delta rebalancing needs into gamma, speed, and charm components:

$$\Delta_{t_1} - \Delta_{t_0} \approx S \times \Gamma \times r_{t_0, t_1} + 0.5 S^2 \times \frac{\partial^2 \Delta}{\partial S^2} \times r_{t_0, t_1}^2 + \frac{\partial \Delta}{\partial t} (t_1 - t_0), \quad (\text{IA.1.1})$$

where the first term quantifies the direct gamma effect, the second term quantifies the effect of gamma change, called “speed,” and the third term quantifies delta time-decay, called “charm.” For small time intervals, it is customary to neglect second-order and time effects, and we also concentrate on the gamma effect in the main analysis. In dollar terms, by multiplying both sides of the equation above by the underlying price, the excess portfolio delta to be offset through rebalancing can be approximated as follows:

$$\widehat{\$ \Delta}_{t_1} \approx S^2 \times \Gamma \times r_{t_0, t_1} = \$ \Gamma \times r_{t_0, t_1} + 0.5 \$Speed \times r_{t_0, t_1}^2 + \$Charm \times (t_1 - t_0), \quad (\text{IA.1.2})$$

where we define dollar speed $\$Speed = S^3 \times \frac{\partial^2 \Delta}{\partial S^2}$ and dollar charm $\$Charm = S \times \frac{\partial \Delta}{\partial t}$.

Empirically, for the 01/2021 to 14/6/2023, for which we can compute the open interest of market makers, both dollar speed and charm are positive on average (and for most days), with charm being very small in magnitude (we compute it for a 10-minute intraday interval). Being positive, both speed and charm will increase excess delta; in other words, they would increase selling pressure on the market, especially after large intraday return realizations (due to the r_{t_0, t_1}^2 term accompanying dollar speed). Large dollar speed and negative return realizations can propagate intraday volatility in case market makers’ open interest gamma is small and overcompensated by the dollar speed term. We estimate a regression as in (10) in the main text, but with dollar speed and charm effects:

$$\begin{aligned} \ln RV_{t+1}^{day} = & b_0 + OI_{MM, t, 0DTE}^{\Gamma} \times (b_1 + b_2 \theta_t) + OI_{MM, t, 0DTE}^{\$Speed} \times (b_3 + b_4 \theta_t) \\ & + OI_{MM, t, 0DTE}^{\$Charm} \times (b_5 + b_6 \theta_t) + \mathbf{CX} + \varepsilon_z, \end{aligned} \quad (\text{IA.1.3})$$

	$\ln RV_{t+1}$			
	(1)	(2)	(3)	(4)
$0DTE\ net\Gamma_t$	-0.075*** (0.018)	-0.074*** (0.018)	-0.062*** (0.019)	-0.070*** (0.018)
$0DTE\ Speed_t$	-0.013*** (0.004)	-0.072 (0.052)	-0.052 (0.033)	
$0DTE\ Charm_t$	0.010 (0.007)	-0.003 (0.010)	0.010 (0.009)	
$0DTE\ net\Gamma_t \times High\ RV_t$		-0.003 (0.024)		
$0DTE\ Speed_t \times High\ RV_t$		0.061 (0.053)		
$0DTE\ Charm_t \times High\ RV_t$		0.026* (0.013)		
$0DTE\ net\Gamma_t \times Neg\ Ret_t$			-0.012 (0.018)	
$0DTE\ Speed_t \times Neg\ Ret_t$			0.042 (0.034)	
$0DTE\ Charm_t \times Neg\ Ret_t$			-0.004 (0.012)	
$0DTE\ Flow\Delta_t$				-0.003 (0.007)
$0DTE\ Flow\Delta_t \times SignRet_t$				0.003 (0.007)
$High\ RV_t$		0.446*** (0.103)		
$Neg\ Ret_t$			-0.105* (0.061)	
$SignRet_t$				0.054* (0.030)
Additional Controls	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Year $\times$ Week-day $\times$ Time FE	Yes	Yes	Yes	Yes
R-squared Adj.	0.832	0.835	0.833	0.833
Obs.	4,899	4,899	4,899	4,899

Table IA.1: Accounting for Speed, Charm, and Initial Hedge. This table reports the results of a regression of log of intraday variance over 30-minutes, following every 30-minute time point, on the market maker SPXW 0DTE open interest gamma, speed and charm, denoted $0DTE\ net\Gamma_t$, $0DTE\ Speed_t$, and $0DTE\ Charm_t$. Each variable interacts with a dummy variable indicating a negative last 30-minute return of the index ($Neg\ Ret_t$) or high (above its sample median) last 30-minute variance of the index ($High\ RV_t$). In the last two specifications, we include $0DTE\ Flow\Delta_t$ and its interaction with the sign of the last 30-minute return of the index ($SignRet_t$). The dependent and continuous independent variables before interaction are scaled to unit variance. The regression controls for three lags of the dependent variable and of the returns and the first difference of volume computed over the same intervals as the lags of the dependent variable. All the continuous controls are interacted with $Neg\ Ret_t$, $High\ RV_t$, and $SignRet_t$. The coefficients of all these controls are suppressed for brevity. Standard errors are based on Newey and West (1987) with five lags. The sample period is from 01/2021 to 14/06/2023.

where θ_t is one of the dummy variables— $High\ RV_t$ indicating past realized 30-minute variance above its sample median, and $Neg\ Ret_t$ indicating negative last 30-minute return of the underlying index. The results in Table IA.1 indicate that, on average, the dollar speed and charm

have mostly an insignificant effect on the future variance. What matters most, however, is that all interactions with a negative return dummy are insignificant, so neither of the excess open interest delta drivers contributes to the "bad" variance propagation. We also find no significant effect of the delta of intraday 0DTE flow interacted with the realized return sign.

IA.1.2 Daily 0DTE Trading Volume Effects

To understand why our finding of no apparent propagation of the underlying index volatility through the 0DTEs market differs from the positive effect of 0DTE trading volume on the underlying variance documented in Brogaard, Han, and Won (2023), we note that our approaches are very different. While Brogaard, Han, and Won directly relate daily variance to the 0DTE volume, we estimate intraday propagation of shocks in a system with absolute returns and trading volumes in 0DTEs and underlying, respectively, and only then relate the intensity of shock propagation to 0DTE volume and other variables. Moreover, in other tests, we focus on whether the potential delta-hedging intensity captured by 0DTE gamma, instead of 0DTE trading volume, propagates recently realized underlying variance and find that it does not.

We dig deeper into the sources of the somewhat conflicting findings using a specification similar to Brogaard, Han, and Won's baseline result. We regress day d intraday variance RV_d of the SPX index on the morning open interest dollar gamma $OI_d^{\$ \Gamma}$, lagged 0DTE volume, for which we use either log of 0DTE dollar volume (with and without delta adjustment), denoted $0DTE\ Volume_{d-1}$, or the proportion of 0DTE dollar trading volume relative to that of all options for the same underlying maturing within the next month, denoted $DTE0\%$.²⁴ Because trading volumes and daily variances can be persistent, we estimate the specification with and without the following controls: lagged values of the dependent variable, which accounts for the persistence of the outcome, and year-fixed effects to account for common trends. The results provided in Table IA.2 are consistent with our inferences after accounting for the above-mentioned controls; importantly, lagged 0DTE volume loses significance in most specifications with added controls and even turns negative and (borderline) significant in Panel B, columns (4) and (8).

²⁴We use options with roots SPX/SPXW and SPY.

	$Vol^{\$}$				$Vol^{\$ \Delta}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. 0DTE Volume as Explanatory Variable</i>								
$OI_{d-1}^{\$ \Gamma}$	-0.278*** (0.065)	-0.201*** (0.047)	-0.020 (0.015)	-0.015 (0.015)	-0.283*** (0.066)	-0.210*** (0.051)	-0.021 (0.015)	-0.019 (0.016)
$0DTE\ Volume_{d-1}$	0.237*** (0.032)	0.452*** (0.091)	0.023 (0.017)	0.058 (0.063)	0.232*** (0.032)	0.410*** (0.091)	0.024 (0.017)	0.068 (0.061)
RV_{d-1}^{day}			0.489*** (0.150)	0.486*** (0.150)			0.489*** (0.150)	0.485*** (0.150)
RV_{d-2}^{day}			0.487*** (0.158)	0.479*** (0.157)			0.487*** (0.158)	0.480*** (0.157)
RV_{d-3}^{day}			0.076 (0.099)	0.078 (0.099)			0.075 (0.099)	0.078 (0.099)
RV_{d-4}^{day}			-0.029 (0.072)	-0.034 (0.071)			-0.029 (0.071)	-0.033 (0.070)
RV_{d-5}^{day}			-0.096 (0.069)	-0.098 (0.069)			-0.096 (0.069)	-0.098 (0.069)
Year Dummies	No	Yes	No	Yes	No	Yes	No	Yes
R-squared Adj.	0.067	0.137	0.726	0.725	0.064	0.137	0.726	0.725
Obs.	1,685	1,685	1,684	1,684	1,685	1,685	1,684	1,684
<i>Panel B. 0DTE% as Explanatory Variable</i>								
$OI_{d-1}^{\$ \Gamma}$	-0.244*** (0.058)	-0.131*** (0.031)	-0.013 (0.013)	-0.003 (0.012)	-0.233*** (0.049)	-0.095*** (0.025)	-0.008 (0.014)	0.008 (0.013)
$0DTE\ Volume_{d-1}$	0.163*** (0.021)	-0.232 (0.185)	0.009 (0.016)	-0.066* (0.037)	0.101*** (0.028)	-0.259 (0.158)	-0.003 (0.016)	-0.078** (0.032)
RV_{d-1}^{day}			0.492*** (0.149)	0.489*** (0.148)			0.493*** (0.149)	0.486*** (0.148)
RV_{d-2}^{day}			0.488*** (0.157)	0.478*** (0.157)			0.488*** (0.157)	0.477*** (0.157)
RV_{d-3}^{day}			0.074 (0.098)	0.078 (0.098)			0.073 (0.097)	0.082 (0.098)
RV_{d-4}^{day}			-0.027 (0.072)	-0.033 (0.071)			-0.026 (0.072)	-0.034 (0.071)
RV_{d-5}^{day}			-0.096 (0.069)	-0.100 (0.069)			-0.096 (0.069)	-0.101 (0.069)
Year Dummies	No	Yes	No	Yes	No	Yes	No	Yes
R-squared Adj.	0.046	0.132	0.725	0.725	0.034	0.137	0.725	0.726
Obs.	1,685	1,685	1,684	1,684	1,685	1,685	1,684	1,684

Table IA.2: Daily Realized Variance vs. 0DTE Trading Volume. This table reports the results of a daily time series regression of intraday variance of the SPX index on the lagged values of open interest dollar gamma $OI_{d-1}^{\$ \Gamma}$ and the lagged 0DTE volume proxy, computed from either dollar or dollar delta volume, as indicated in the Table headers. In Panel A, $0DTE\ Volume_{d-1}$ is the log of the volume variable indicated in the Table header. In Panel B, $0DTE\ \%_{d-1}$ is the proportion of 0DTE trading volume indicated in the Table header relative to the total of the corresponding trading volume of all options (with roots SPY, SPX, and SPXW) expiring within the next month. As additional controls, we use lagged intraday variances and year fixed effects. All continuous variables are scaled to unit variance. Newey and West (1987) standard errors with five lags are reported in parentheses. The sample period is from 01/2012 to 30/04/2024.

IA.1.3 Start-of-Day Open Interest and Daily Variance

We also analyze the impact of the total open interest on subsequent volatility. This analysis is motivated by the possibility that some traders, not necessarily market makers, delta hedge their inventory. One can hypothesize that short gamma positions have a stronger motive for maintaining a dynamic hedge because negative convexity leads to losses for short gamma positions in the case of large market moves. An assumption we make here is that the total open interest gamma is highly correlated to the net gamma of the delta hedgers. Thus, we project future variance onto total 0DTE open interest gamma at the market open to establish whether morning gamma is associated with higher or lower subsequent intraday volatility.

For the test we use log realized variance $\ln RV_d^{day}$ computed from equation (3) using all intraday 1-minute intervals for a day, the start-of-day open interest cash gamma $OI_d^{\$ \Gamma}$ computed in equation (4) and aggregated for the pre-defined DTE buckets for SPX (only longer than 0 DTE), SPXW and SPY options, to capture total rebalancing risk due to fluctuations in the underlying market index. We estimate the following regression:

$$\ln RV_d^{day} = b_0 + OI_{d:sod,dte}^{\$ \Gamma} \times (b_1 + b_2 \ln RV_d^{on} + b_3 \ln RV_{d-1}^{day}) + \mathbf{CX} + \varepsilon_d, \quad (\text{IA.1.4})$$

where $\mathbf{X}$ is a vector of controls including overnight variance RV_d^{on} , five lags of intraday variance RV_d^{day} (i.e., lags of the dependent variable), and year dummies.²⁵ We standardize all continuous variables before an interaction to unit variance. Table IA.3 provides no indication that total risk in 0DTEs is associated with the propagation of the overnight and lagged intraday variances to the subsequent daily variance. The coefficients on the interaction between the gamma levels and variances are insignificant for 0DTEs. We have weak evidence for the other DTE buckets that high gamma and overnight variance interaction terms are linked to a lower variance the next day. Moreover, open interest gamma in the morning of the day d is mostly *negatively* related to the intraday volatility, and the coefficient is significant for intermediate maturity buckets.

²⁵Including more lags of overnight variance for interactions has no effect on magnitude and significance of other coefficients, so we kept only one. Excluding the interaction of gamma with lagged intraday variance does not materially change the results. In Robustness Section 5 we also estimate a similar regression without interactions, including lagged daily trading volume in 0DTEs and lagged start-of-day open interest gamma. The volume coefficient turns insignificant once we control for lags of the dependent variable.

	$\ln RV_d^{day}$				
	0-DTE	1-DTE	2-5 DTE	6-10 DTE	11-22 DTE
$OI\Gamma_d \times \ln RV_d^{on}$	-0.007 (0.018)	-0.012 (0.017)	0.004 (0.011)	-0.015 (0.011)	-0.019 (0.014)
$OI\Gamma_d \times \ln RV_{d-1}^{day}$	-0.020 (0.017)	-0.009 (0.020)	-0.000 (0.018)	-0.034** (0.017)	-0.009 (0.021)
$OI\Gamma_d$	-0.264 (0.169)	-0.138 (0.186)	-0.070 (0.188)	-0.456*** (0.176)	-0.155 (0.206)
$\ln RV_d^{on}$	0.047 (0.033)	0.044 (0.028)	0.035* (0.019)	0.062*** (0.018)	0.069*** (0.022)
$\ln RV_{d-1}^{day}$	0.362*** (0.047)	0.177*** (0.042)	0.148*** (0.038)	0.204*** (0.037)	0.169*** (0.040)
Year $\times$ Week FE	Yes	Yes	Yes	Yes	Yes
R-squared Adj.	0.796	0.806	0.787	0.784	0.783
Obs.	1,601	1,605	2,829	2,937	2,969

Table IA.3: Volatility Propagation by Open Interest Gamma. This table reports the results of a regression of log of intraday variance on the level of overnight variance and lagged intraday variance both interacted with start-of-day open interest dollar gamma by DTE buckets, using the specification in (IA.1.4). Start-of-day open interest is converted to dollar gamma $OI_{d:sod,dte}^{\$ \Gamma}$ at 10:00 each day and aggregated for SPY, SPXW, and SPX (only for non-0DTE) options with moneyness levels in $[0.5, 1.5]$ for each DTE bucket. Realized variances are computed from intraday and overnight returns for the SPX index. Before the interaction, dependent and continuous independent variables are scaled to unit variance. Five lags of the dependent variable are included, but only one is shown for brevity. Standard errors in parentheses use Newey and West (1987) with five lags. The sample period is from 01/2012 to 30/04/2024.

Overall, we do not find evidence that higher open interest gamma levels destabilize prices and contribute to clustered volatility shocks.

IA.1.4 VAR Analysis Extensions

Impulse Response Functions Dynamics. In the main text, we made an informal statistical inference about the significance of the impulse response intensity time dynamics based on the overlap of the confidence bounds. To see whether the propagation of the shocks to volatility and both 0DTE and underlying trading volumes is related to 0DTE trading activity, we relate each day's cumulative generalized impulse response after five time steps (i.e., five minutes for the 1-minute VAR frequency) to year dummies, overnight and intraday variances, and to dummy variables for considerable (more than one standard deviation) jumps of open interest dollar gamma and 0DTEs trading volume relative their past averages.²⁶ Table IA.4 shows the results of the regressions. Neither high open interest gamma (*High* $OI_{dte,d}^{\$ \Gamma}$) nor high 0DTE trading

²⁶Both standard deviation and averages are computed from the past 21 daily observations.

Impulse	Variance			0DTE Vol.			Und.Vol.		
Propagation	Variance	0DTE Vol.	Und.Vol.	Variance	0DTE Vol.	Und.Vol.	Variance	0DTE Vol.	Und.Vol.
$High\ OI_d^{\$F}$	0.014 (0.022)	-0.014** (0.006)	-0.010 (0.006)	-0.008 (0.015)	0.005 (0.006)	-0.005 (0.006)	-0.015 (0.022)	-0.006 (0.007)	-0.003 (0.007)
$High\ \ln Vol_d^{\$ \Delta}$	0.021 (0.019)	0.012*** (0.004)	0.013** (0.005)	0.008 (0.014)	0.012** (0.005)	0.003 (0.005)	-0.005 (0.018)	0.009 (0.006)	0.010** (0.005)
$\ln RV_d^{on}$	-0.023*** (0.005)	-0.004*** (0.001)	-0.004*** (0.001)	-0.010*** (0.003)	-0.000 (0.001)	-0.004*** (0.001)	-0.020*** (0.004)	-0.004*** (0.001)	-0.003** (0.001)
$\ln RV_d^{day}$	0.011 (0.008)	0.009*** (0.002)	0.006*** (0.002)	0.006 (0.005)	0.002 (0.002)	-0.002 (0.002)	0.006 (0.008)	0.004* (0.002)	-0.003 (0.002)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Lags	5	5	5	5	5	5	5	5	5
R-squared Adj.	0.042	0.074	0.039	0.339	0.121	0.444	0.087	0.482	0.097
Obs.	1,679	1,679	1,679	1,679	1,679	1,679	1,679	1,679	1,679

Table IA.4: Conditional Generalized Impulse Response Functions. This table reports the analysis of the 5-step cumulative responses based on the cumulative gIRFs from the VAR system (11) estimated daily using 1-minute data for Variance, 0DTE Volume (0DTE Vol., in dollar delta terms), and Underlying Volume (Und.Vol., in dollar terms). We regress the cumulative gIRFs on its five lags, log of overnight variance ($\ln RV_d^{on}$), log of intraday variance ($\ln RV_d^{day}$), and dummy variables $High\ OI_d^{\$F}$ and $High\ \ln Vol_d^{\$ \Delta}$, equal one for large positive deviations of day d values from their rolling-window averages. Newey and West (1987) standard errors with five lags are reported in parentheses. The sample period is from 01/2012 to 30/04/2024.

volume ($High\ \ln Vol_d^{\$ \Delta}$) is linked to significantly higher responses to shocks in the system, except for one case, where the propagation of 0DTE volume to the underlying market volume is significantly larger when 0DTE volume is high relative to its recent past average.

VAR Analysis for Other DTE Buckets. To see whether the time trend in impulse response intensity is specific to the 0DTEs, we re-estimate the VAR system in (11) using option volume variable for other maturity buckets and also for the volume aggregated over all maturities up to 22 trading days. Figure IA.1 shows the time-series plots of the smoothed cumulative responses five steps after a shock. In Panel A, we observe a stronger volatility response to short-term options' trading volume for all years and its very pronounced upward trend starting around the second quarter of 2020, led by 0- and 1DTEs. Options with more than a week to expiration have a stable or even a slightly decreasing (for 11-22DTEs) cumulative response. The responses are also sizeable, reaching almost 0.4 of the standard deviation. Realized return (volatility) shocks also propagate stronger to short-term options but with a negligible cumulative effect size. Option and underlying trading look very similar, with a relatively sizeable shock propagation in both directions and an increasing trend for short-term options. Notably, the effects (in both directions) for the *aggregate* options volume seem to be almost entirely driven by ultra-short-term buckets. A possible explanation for the observed time effects is an increasing integration

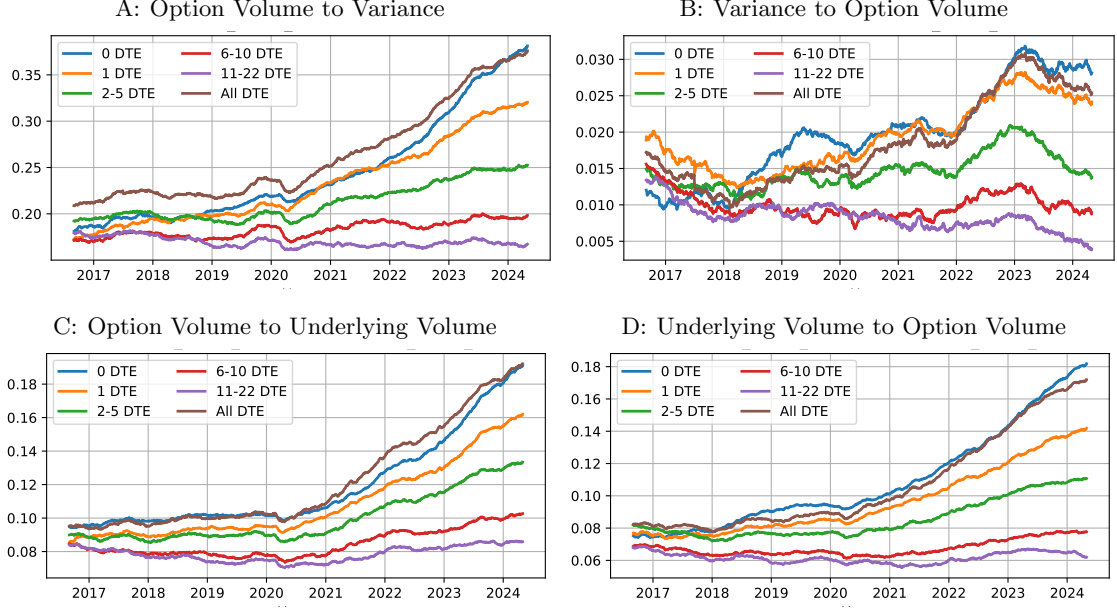


Figure IA.1: Dynamics of Option Volume and Variance Cumulative gIRFs. The figure shows the smoothed (exponential moving average with a half-life of 252 days) time series of the cumulative generalized impulse response functions after five steps for the VAR system in (11) estimated each day for the 1-minute frequency with $n = 5$ lags, separately for options volume in different maturity buckets. The variables are scaled daily to unit variance, and the response is calculated for one standard deviation shock to a given variable. The sample period is from 01/2012 to 30/04/2024.

of the underlying and options markets for liquid contracts (e.g., Dew-Becker and Giglio 2023), which is consistent with the increased correlation between trading volumes (from relatively uniform annual average levels of 0.25-0.3 before 2021 to 0.38, 0.44, and 0.59 in the next three years, respectively). At the same time, trading in both 0DTE and underlying markets became much smoother over the years, with the average daily standard deviation of intraday log volume differences dropping from 1.49 in 2012 to 0.49 in 2023 for 0DTEs and from 0.80 to 0.47 for the underlying instruments.